

stat_gformula_composition_minimax (transition_smoothness_elbow)

TL;DR. For the two-decision g-formula target $\psi = E[Y(1, 1)]$, the note proves a computable cross-fitted first-order upper bound and three converse floors. The honest bracket is

$$\max\{n^{-1/2}, n^{-r_1}, n^{-r_2}\} \lesssim \mathfrak{R}_n(\mathcal{M}) \lesssim n^{-1/2} + n^{-(\gamma_1 + \gamma_t)} + n^{-(\gamma_2 + \gamma_b)},$$

under the stated nonemptiness and converse-slack qualifications. The incoming lower channel pairs the first propensity with incoming-state transition smoothness; the terminal channel is the familiar single-time-type component. The auxiliary monotone-MAR comparator is now a complete statistical law, including its treated-outcome kernel, but remains distinct from the causal full-data extension. The bounds do not match in general, so neither the minimax rate nor a root- n if-and-only-if elbow is claimed.

1 Project justification

Gap. Optimal non-root- n theory for time-varying treatment effects remains open. In this model, the unresolved mathematical issue is the gap between the transition-mediated converse exponent $r_1 = 2(\alpha_1 + t)/\{2(\alpha_1 + t) + d\}$ and the first-order achievable exponent $\gamma_1 + \gamma_t$, together with the analogous terminal-channel gap.

Niche. The note isolates an incoming-transition channel with no nonconstant counterpart when the transition is state independent. It uses incoming $C^t(L^1)$ regularity and permits a fixed subclass with genuinely rough outgoing-state densities. The public comparison is made on a fully specified auxiliary monotone-MAR coarsening, including an outcome drawn from the longitudinal treated-outcome regular conditional kernel.

Fill. The contribution is an explicit nonmatching bracket: a sample-measurable cross-fitted first-order estimator, a chained fuzzy lower construction pairing π_1 with g , the standard terminal fuzzy channel, and a parametric floor. On the fixed rough subclass, the auxiliary law draws $(L_0, L_1, Y^{\text{MAR}})$ from $p_1 g K_P$, then applies monotone response indicators with probabilities $\pi_1(L_0)$ and $\pi_2(L_1)$. Its observed-data coarsening matches the longitudinal treated-regime coordinates without equating L_1 with $H_2(1)$.

2 Related work

Longitudinal g-formula identification is standard, and the terminal lower channel parallels single-time missing-data and treatment-effect constructions, includ-

ing Robins, Li, Tchetgen Tchetgen, and van der Vaart. The incoming channel instead perturbs the first propensity and treated transition jointly because $m_1 = T_g m_2$ is not variation independent of g . For the public comparator, Section 7.2.2 and Theorem 49 of Robins et al. require a positive-order Hölder joint density for (L_0, L_1) . The completed auxiliary monotone-MAR law has full data $(L_0, L_1, Y^{\text{MAR}})$, response indicators satisfying the displayed MAR identities, and observed coarsening $(R_0, L_0, R_0 L_1, R_1, R_0 R_1 Y^{\text{MAR}})$. For laws induced from $\mathcal{M}_{\text{rough}}$, its joint state density $p_1 g = g_0 + r$ has a nonremovable outgoing-state jump, so those auxiliary laws are not covered by the stated condition. This is only noncoverage: no reverse inclusion, broad non-overlap, or priority claim is made. The note gives a model-specific partial answer to the open non-root- n agenda described by Cinelli et al. The Bonvini–Kennedy–Keele item remains disclosed only as unavailable work in progress; no assumptions, rates, inclusion relation, non-overlap, or priority inference is drawn from it.

3 Setup

Target (estimand / structural parameter): $\psi = E_{\tilde{P}}[Y(1, 1)]$. Identifying functional / recovery map: $\psi = E_{Q_2}[m_2(H_2)] = E_{Q_1}[\int m_2(h_2)g(h_2 | H_1) dh_2]$.

- n — sample size; $\mathbb{N}$; index := Number of observed trajectories.
- d — state dimension; $\mathbb{N}$; index := Dimension of each state.
- O_i — observed trajectory; $[0, 1]^d \times \{0, 1\} \times [0, 1]^d \times \{0, 1\} \times [-1, 1]$; data := $O_i = (H_{1i}, A_{1i}, H_{2i}, A_{2i}, Y_i)$.
- P — observed-data law; probability laws; model := Law of one observed trajectory.
- $\tilde{P}$ — full-data law; probability laws; model := Full-data extension of P .
- H_1 — first state; $[0, 1]^d$; data := State before the first decision.
- H_2 — second state; $[0, 1]^d$; data := State before the second decision.
- A_1 — first decision; $\{0, 1\}$; data := First binary treatment.
- A_2 — second decision; $\{0, 1\}$; data := Second binary treatment.
- Y — outcome; $[-1, 1]$; data := Observed terminal outcome.
- $Y(a_1, a_2)$ — potential outcome; $[-1, 1]$; causal object := Outcome under $a_1, a_2 \in \{0, 1\}$.
- $H_2(a_1)$ — potential state; $[0, 1]^d$; causal object := Second state under a_1 .
- h_1 — argument; $[0, 1]^d$; argument := Generic value of H_1 .

- h_2 — argument; $[0, 1]^d$; argument := Generic value of H_2 .
- π_1 — propensity; $[0, 1]^d \rightarrow (0, 1)$; nuisance := $P(A_1 = 1 \mid H_1 = h_1)$.
- π_2 — propensity; $[0, 1]^d \rightarrow (0, 1)$; nuisance := $P(A_2 = 1 \mid H_2 = h_2, A_1 = 1)$.
- g — transition density; $(h_2, h_1) \mapsto [0, \infty)$; nuisance := Density of $H_2 \mid (H_1 = h_1, A_1 = 1)$.
- m_2 — terminal regression; $[0, 1]^d \rightarrow [-1, 1]$; nuisance := $E_P[Y \mid H_2 = h_2, A_1 = A_2 = 1]$.
- m_1 — backward regression; $[0, 1]^d \rightarrow [-1, 1]$; nuisance := $\int m_2(h_2)g(h_2 \mid h_1) dh_2$.
- Q_1 — interventional law; probability measures; integration measure := P_{H_1} .
- Q_2 — interventional law; probability measures; integration measure := $Q_2(dh_2) = \int P_{H_1}(dh_1)g(h_2 \mid h_1) dh_1$.
- ψ — causal estimand; $\mathbb{R}$; target := $E_{\bar{P}}[Y(1, 1)]$.
- α_1 — smoothness; $(0, \infty)$; parameter := Hölder order of π_1 .
- α_2 — smoothness; $(0, \infty)$; parameter := Hölder order of π_2 .
- β — smoothness; $(0, \infty)$; parameter := Hölder order of m_2 .
- t — smoothness; $(0, \infty)$; parameter := Incoming-state transition Hölder order.
- M — radius; $(0, \infty)$; parameter := Regularity-ball radius.
- c — overlap constant; $(0, 1/2)$; parameter := Positivity constant.
- $\underline{q}$ — density bound; $(0, \infty)$; parameter := Lower design-density bound.
- $\bar{q}$ — density bound; $(0, \infty)$; parameter := Upper design-density bound.
- $\mathcal{H}_d^s(M)$ — Hölder ball; function classes; class := Class in *def : holder - ball*.
- $\mathcal{G}_d^t(M)$ — transition class; density classes; class := Class in *def : transition - ball*.
- $\mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c)$ — model; law classes; class := Class in *def : longitudinal - model*.
- $\mathcal{M}$ — model shorthand; law classes; class := $\mathcal{M} = \mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c)$.
- s_1 — channel index; $(0, \infty)$; rate notation := $\alpha_1 + t$.

- s_2 — channel index; $(0, \infty)$; rate notation $:= \alpha_2 + \beta$.
- r_1 — rate exponent; $(0, 1)$; rate notation $:= 2s_1/(2s_1 + d)$.
- r_2 — rate exponent; $(0, 1)$; rate notation $:= 2s_2/(2s_2 + d)$.
- $\hat{\psi}$ — estimator; $\mathbb{R}$; estimation object $:=$ Sample-measurable estimator in the minimax definition.
- $\hat{\psi}^{\text{SDR}}$ — estimator; $\mathbb{R}$; estimation object $:=$ Direct-backward estimator in *def : direct – backward – estimator*.
- $\hat{\psi}^{\text{HOIF}}$ — estimator; $\mathbb{R}$; estimation object $:=$ Corrected estimator in *def : hoif – chains*.
- $\mathfrak{R}_n$ — minimax RMSE; $[0, \infty)$; frontier $:=$ Risk in *def : minimax – rmse*.
- $\mathcal{T}, \mathcal{B}, \mathcal{G}, \mathcal{E}$ — folds; disjoint index sets; fold provenance $:=$ Nuisance, backward, Gram, and evaluation folds.
- $\mathcal{F}_T, \mathcal{F}_{TB}, \mathcal{F}_{TBG}$ — sigma-fields; sigma-fields; fold provenance $:=$ Sigma-fields generated respectively by $\mathcal{T}$, $\mathcal{T} \cup \mathcal{B}$, and $\mathcal{T} \cup \mathcal{B} \cup \mathcal{G}$.
- $\hat{\pi}_1, \hat{\pi}_2, \hat{m}_2$ — nuisance fits; clipped functions; estimation input $:=$ Fits trained on $\mathcal{T}$.
- $\hat{D}_2$ — pseudo-outcome; $\mathbb{R}$; sequential object $:=$ Augmented second-stage outcome in *def : direct – backward – estimator*.
- $\hat{m}_1$ — backward fit; functions; estimation input $:=$ Wavelet regression fit trained on $\mathcal{B}$.
- $b_1^\dagger$ — conditional target; functions; sequential object $:=$ Conditional target in *def : direct – backward – estimator*.
- w_1, w_2, e_2, T_g — remainder notation; functions and operator; analysis object $:=$ Objects in *def : remainder – notation*.
- B_1, B_2, B_T — bias components; $\mathbb{R}$; analysis object $:=$ Components in *lem : direct – bias*.
- $Z_j, K_j, \hat{\Omega}_{j,b}, K_{j,b}$ — projection objects; wavelets, matrices, and kernels; projection object $:=$ Objects in *def : blocked – projection*.
- K_1 — projection kernel; kernels; projection object $:=$ First-state blocked projection kernel.
- K_2 — projection kernel; kernels; projection object $:=$ Second-state blocked projection kernel.
- B_j, κ_j — block parameters; $\mathbb{N} \times \mathbb{N}$; projection object $:=$ Number and maximum size of wavelet blocks.

- $S_1, R_1, T_1, S_2, \tilde{R}_2, T_2$ — observable residuals; $\mathbb{R}$; HOIF factor := Residual factors in *def : hoif – chains*.
- $\Delta_{1,k}, \Delta_{2,k}, \Delta_{T,p,r}$ — U-statistic corrections; $\mathbb{R}$; estimator := Symmetrized corrections in *def : hoif – chains*.
- $U_{\mathcal{E}}^q$ — U-statistic operator; symmetric kernels to reals; estimator := Evaluation-fold U-statistic operator in *def : hoif – chains*.
- $\mathcal{I}_T, q_1, q_2$ — truncation indices; finite index sets and integers; order calibration := Orders selected in *def : hoif – chains*.
- a_1, a_2, b, b_1 — preliminary rates; positive sequences; order calibration := Rates in *def : preliminary – rates*.
- $u_{j,h}, v, v_{j,h}, J, h, z, \lambda, \eta, \epsilon, \delta, a, b_0$ — fuzzy-experiment and rough-subclass objects; functions, indices, signs, amplitudes, likelihood channels, and a fixed model subclass; lower-bound construction := In the rough construction, $v(h_2) = \cos(4\pi h_{2,1})$, and $p_1 \equiv 1$ denotes the uniform density of H_1 . The remaining objects belong to the two fuzzy experiments and the rough comparator construction, including $q, a, g_0, g_{z,\lambda}^{\text{rough}}, \mathcal{M}_{\text{rough}}, F_j, E_j$.
- $\nu_+, \nu_-, \bar{P}_+, \bar{P}_-$ — mixtures; probability measures; lower-bound construction := Mixing and product-mixture laws in the fuzzy experiments.
- H^2 — squared Hellinger distance; $[0, 1]$; divergence := Squared Hellinger distance between two probability laws.
- ρ — affinity constant; $(0, 1)$; lower-bound construction := Uniform positive lower bound for product-mixture affinity.
- ϑ — parametric coordinate; $[-1/4, 1/4]$; lower-bound construction := Coordinate of the regular parametric submodel.
- $\mu_{a_1 a_2}$ — potential-outcome mean; functions; lower-bound construction := Conditional mean used in the full-data fuzzy construction.
- $U_{A1}, U_{H2}, U_{A2}, U_Y^{a_1 a_2}$ — exogenous uniforms; uniform random variables; causal construction := Independent randomizers for the full-data extensions.
- $t_0, \theta_0, w, v_0, \chi, \xi, \mathcal{G}_{\text{AH}}^{t_0}(M, \theta_0)$ — separation objects; parameters, functions, and a class; separation object := Objects in *def : analytic-outgoing-subclass*.

4 Assumptions

Assumption 1 (ass:iid). $O_1, \dots, O_n \stackrel{\text{iid}}{\sim} P$.

Standard (*independent and identically distributed sampling*, [8]).

Assumption 2 (ass:consistency). $Y = Y(A_1, A_2) \quad \tilde{P}\text{-a.s.}$

Standard (*consistency*, [7]).

Assumption 3 (ass:exchangeability-one). $Y(1, 1) \perp_{\tilde{P}} A_1 \mid H_1.$

Standard (*first-period sequential exchangeability*, [7]).

Assumption 4 (ass:exchangeability-two). $Y(1, 1) \perp_{\tilde{P}} A_2 \mid (H_1, H_2, A_1 = 1).$

Standard (*second-period sequential exchangeability*, [7]).

Assumption 5 (ass:overlap-one). $c \leq \pi_1(h_1) \leq 1 - c.$

Standard (*strict positivity*, [7]).

Assumption 6 (ass:overlap-two). $c \leq \pi_2(h_2) \leq 1 - c.$

Standard (*strict positivity*, [7]).

Assumption 7 (ass:outcome-state-sufficiency). $E_P[Y \mid H_1, H_2, A_1 = 1, A_2 = 1] = m_2(H_2).$

Novel. *This is an explicitly conditional theoretical state-sufficiency restriction. It is realistic only when the recorded second state contains the history that determines the terminal conditional mean; no empirical validation or protocol evidence is claimed here.*

Assumption 8 (ass:treatment-state-sufficiency). $P(A_2 = 1 \mid H_1, H_2, A_1 = 1) = \pi_2(H_2).$

Novel. *This is a conditional theoretical restriction appropriate when the second decision rule uses only the recorded second state. Its applicability requires a future protocol to prespecify the state and document the retained history.*

Assumption 9 (ass:holder-pi-one). $\pi_1 \in \mathcal{H}_d^{\alpha_1}(M).$

Standard (*isotropic Hölder smoothness*, [8]).

Assumption 10 (ass:holder-pi-two). $\pi_2 \in \mathcal{H}_d^{\alpha_2}(M).$

Standard (*isotropic Hölder smoothness*, [8]).

Assumption 11 (ass:holder-terminal). $m_2 \in \mathcal{H}_d^{\beta}(M).$

Standard (*isotropic Hölder smoothness*, [8]).

Assumption 12 (ass:transition-holder). $g \in \mathcal{G}_d^t(M).$

Novel. *Conditional transition laws generated by an incoming-state smooth mixture weight with arbitrary bounded outgoing components satisfy this condition. It permits rough outgoing biomarker distributions while requiring stable variation of the full conditional law across the incoming state.*

Assumption 13 (ass:design-density). $0 < \underline{q} \leq dQ_j/dh_j \leq \bar{q} < \infty \quad (j = 1, 2).$

Standard (*bounded random-design density*, [4]).

5 Definitions

Definition 1 (def:holder-ball). $\mathcal{H}_d^s(M) := \mathcal{H}_d^s(M) = \{f : [0, 1]^d \rightarrow \mathbb{R} : \|f\|_{C^s([0,1]^d)} \leq M\}$. — the class of all objects satisfying: .

Definition 2 (def:transition-ball). $\mathcal{G}_d^t(M) := \mathcal{G}_d^t(M) = \{g : g(h_2 | h_1) \geq 0, \int_{[0,1]^d} g(h_2 | h_1) dh_2 = 1, \|h_1 \mapsto g(\cdot | h_1)\|_{C^t([0,1]^d; L^1(dh_2))} \leq M\}$. — the class of all objects satisfying: .

Definition 3 (def:longitudinal-model). $\mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c) := \mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c) = \{P : \exists \tilde{P} \text{ extending } P \text{ satisfying } \text{ass:consistency}, \text{ass:exchangeability-one}, \text{ass:exchangeability-two}, \text{ass:overlap-one}, \text{ass:exchangeability-two}, \text{ass:overlap-one}, \text{ass:overlap-two}, \text{ass:outcome-state-sufficiency}, \text{ass:treatment-state-sufficiency}, \text{ass:holder-pi-one}, \text{ass:holder-pi-two}, \text{ass:holder-terminal}, \text{ass:transition-holder}, \text{ass:design-density}\}$. — the class of all objects satisfying: .

Definition 4 (def:gformula-objects). *g-formula objects* $:= Q_1 = P_{H_1}, Q_2(dh_2) = \int P_{H_1}(dh_1)g(h_2 | h_1) dh_1, m_1(h_1) = \int m_2(h_2)g(h_2 | h_1) dh_2, \psi(P) = E_{Q_1}[m_1(H_1)]$. (inputs: $\backslash(P\backslash), \backslash(g\backslash), \backslash(m\backslash)$)

Definition 5 (def:minimax-rmse). $\mathfrak{R}_n := \mathfrak{R}_n(\mathcal{M}) = \inf_{\hat{\psi}} \sup_{P \in \mathcal{M}} E_P[(\hat{\psi} - \psi(P))^2]^{1/2}$. (inputs: $\backslash(n\backslash), \backslash(\backslash \text{mathcal M}(\backslash \alpha_1, \backslash \alpha_2, \backslash \beta, t; M, c)\backslash), \backslash(\backslash \psi\backslash)$)

Definition 6 (def:remainder-notation). *sequential remainder notation* $:= w_j = \pi_j/\hat{\pi}_j, e_2 = (1 - w_2)(\hat{m}_2 - m_2), T_g f(h_1) = \int f(h_2)g(h_2 | h_1) dh_2$. (inputs: $\backslash(\backslash \pi_1\backslash), \backslash(\backslash \pi_2\backslash), \backslash(\backslash \widehat{\pi}_1\backslash, \backslash \widehat{\pi}_2\backslash, \backslash \widehat{m}\backslash), \backslash(m\backslash), \backslash(g\backslash)$)

Definition 7 (def:blocked-projection). *blocked projection kernels* $:=$ For $j = 1, 2$, take a boundary-corrected wavelet vector Z_{j, K_j} with $K_j \asymp n^{2d/(2s_j+d)}$. Partition its coordinates into $B_j = \lceil K_j/\kappa_j \rceil$ consecutive blocks $Z_{j,b}$ of size at most $\kappa_j = \lfloor |\mathcal{G}|/2 \rfloor$. Define $\hat{\Omega}_{1,b} = \mathbb{P}_{\mathcal{G}}[Z_{1,b}Z_{1,b}^T]$, $\hat{\Omega}_{2,b} = \mathbb{P}_{\mathcal{G}}[A_1 Z_{2,b}Z_{2,b}^T/\hat{\pi}_1(H_1)]$, and $K_{j,b}(x, x') = Z_{j,b}(x)^T \hat{\Omega}_{j,b}^{-1} Z_{j,b}(x')$ when the inverse exists, with $K_{j,b}(x, x') = 0$ otherwise; set $K_j = \sum_{b=1}^{B_j} K_{j,b}$. This total block construction is used whenever $K_j > |\mathcal{G}|$. (inputs: $\backslash(\backslash \text{mathcal G}\backslash), \backslash(\backslash \widehat{\pi}_1\backslash), \backslash(s_1\backslash), \backslash(s_2\backslash)$)

Definition 8 (def:preliminary-rates). *preliminary nuisance rates* $:= a_j = n^{-\alpha_j/(2\alpha_j+d)}$ ($j = 1, 2$), $b = n^{-\beta/(2\beta+d)}$, $b_1 = n^{-t/(2t+d)} + a_2 b$, with $L^2(Q_j)$ errors of orders a_j, b , and b_1 , respectively. (inputs: $\backslash(\backslash \alpha_1\backslash), \backslash(\backslash \alpha_2\backslash), \backslash(\backslash \beta\backslash), \backslash(t\backslash), \backslash(d\backslash)$)

Definition 9 (def:direct-backward-estimator). *direct-backward SDR estimator* $:=$ On $\mathcal{T}$, fit clipped $\hat{\pi}_1, \hat{\pi}_2, \hat{m}_2$. On the independent fold $\mathcal{B}$, form $\hat{D}_2 = \hat{m}_2(H_2) + A_2\{Y - \hat{m}_2(H_2)\}/\hat{\pi}_2(H_2)$ and regress $\hat{D}_2$ on H_1 among $A_1 = 1$ using a boundary-corrected wavelet sieve, obtaining $\hat{m}_1$. Define $b_1^\dagger(h_1) = E_P[\hat{D}_2 | H_1 = h_1, A_1 = 1, \mathcal{F}_T]$. On $\mathcal{E}$, set $\hat{\psi}^{\text{SDR}} = \mathbb{P}_{\mathcal{E}}[\hat{m}_1(H_1) + A_1\{\hat{D}_2 - \hat{m}_1(H_1)\}/\hat{\pi}_1(H_1)]$. (inputs: $\backslash(O\backslash), \backslash(\backslash \text{mathcal T}, \backslash \text{mathcal B}, \backslash \text{mathcal E}\backslash), \backslash(\backslash \widehat{\pi}_1\backslash, \backslash \widehat{\pi}_2\backslash, \backslash \widehat{m}\backslash)$)

Definition 10 (def:hoif-chains). *projected sequential corrections* := On $\mathcal{E}$, let $\omega = A_1/\widehat{\pi}_1(H_1)$, $S_1 = 1 - \omega$, $R_1 = \omega - 1$, $T_1 = \omega(\widehat{D}_2 - \widehat{m}_1(H_1))$, $S_2 = \omega(1 - A_2/\widehat{\pi}_2(H_2))$, $\widetilde{R}_2 = \omega(A_2/\widehat{\pi}_2(H_2) - 1)$, and $T_2 = \omega A_2\{Y - \widehat{m}_2(H_2)\}/\widehat{\pi}_2(H_2)$. With $U_{\mathcal{E}}^q f = (|\mathcal{E}|)_q^{-1} \sum_{i_1 \neq \dots \neq i_q \in \mathcal{E}} f(O_{i_1}, \dots, O_{i_q})$, define $\Delta_{1,k} = (-1)^k U_{\mathcal{E}}^{k+2} \text{Sym}[S_1 K_1 (R_1 K_1)^k T_1]$, $\Delta_{2,k} = (-1)^k U_{\mathcal{E}}^{k+2} \text{Sym}[S_2 K_2 (\widetilde{R}_2 K_2)^k T_2]$, and $\Delta_{T,p,r} = (-1)^{p+r} U_{\mathcal{E}}^{p+r+3} \text{Sym}[S_1 K_1 (R_1 K_1)^p S_2 K_2 (\widetilde{R}_2 K_2)^r T]$ where adjacent kernels join the state recorded by their adjacent residual factors, Sym averages all ordered role assignments, and empty powers equal one. Thus $\Delta_{T,0,0}$ has coefficient +1. Define $q_1 = \min\{q \geq 0 : a_1^{q+1} b_1 = o(n^{-r_1})\}$, $q_2 = \min\{q \geq 0 : a_2^{q+1} b = o(n^{-r_2})\}$, choose a finite multi-index set $\mathcal{I}_T \subset \mathbb{N}_0^2$ satisfying $\sum_{(p,r) \notin \mathcal{I}_T} a_1^{p+1} a_2^{r+1} b = o(n^{-\min\{r_1, r_2\}})$, and set $\widehat{\psi}^{\text{HOIF}} = \widehat{\psi}^{\text{SDR}} + \sum_{k=0}^{q_1} \Delta_{1,k} + \sum_{k=0}^{q_2} \Delta_{2,k} + \sum_{(p,r) \in \mathcal{I}_T} \Delta_{T,p,r}$. (inputs: $\backslash(\backslash\widehat{D_2}\backslash)$, $\backslash(\backslash\widehat{m_1}\backslash)$, $\backslash(\backslash\widehat{m_2}\backslash)$, $\backslash(\backslash\widehat{pi_1}\backslash, \backslash\widehat{pi_2}\backslash)$, $\backslash(K_1\backslash)$, $\backslash(K_2\backslash)$, $\backslash(a_1, a_2, b, b_1\backslash)$, $\backslash(r_1\backslash)$, $\backslash(r_2\backslash)$)

Definition 11 (def:parametric-submodel). *regular parametric submodel* := For every $\vartheta \in [-1/4, 1/4]$, let P_{ϑ} be the observed-data law with $H_1 \sim \text{Unif}([0, 1]^d)$, $P_{\vartheta}(A_1 = 1 \mid H_1) = 1/2$, $H_2 \mid (H_1, A_1) \sim \text{Unif}([0, 1]^d)$, and $P_{\vartheta}(A_2 = 1 \mid H_1, H_2, A_1) = 1/2$. Thus $\pi_1 = \pi_2 = 1/2$ and $g(h_2 \mid h_1) = 1$. Set $P_{\vartheta}(Y = 1 \mid H_2, A_1 = 1, A_2 = 1) = (1 + \vartheta)/2$, and set every off-regime conditional outcome law to the fixed fair Rademacher law. The full-data extension uses independent uniforms and the same potential-outcome construction as the fuzzy experiments. (inputs: $\backslash(H_1\backslash)$, $\backslash(H_2\backslash)$, $\backslash(A_1\backslash)$, $\backslash(A_2\backslash)$, $\backslash(Y\backslash)$)

Definition 12 (def:first-fuzzy-experiment). *period-one paired fuzzy experiment and rough comparator subclass* := Choose a nonzero C^∞ reflection-odd bump u supported on a symmetric cube, with disjoint copies $u_{j,h}$, $J \asymp h^{-d}$, signs $z, \lambda \in \{-1, 1\}^J$, $\eta = c_\eta h^{\alpha_1}$, and $\epsilon = c_\epsilon h^t$. Put $\pi_{1,z} = 1/2 + \eta \sum_j z_j u_{j,h}$, $\pi_2 = 1/2$, and $m_2 = \delta v$. The flat-base family used by *thm:transition-mediated-converse-first-order-bracket* is

$$g_{z,\lambda}^{\text{flat}}(h_2 \mid h_1) = 1 + \epsilon \sum_j \frac{\lambda_j u_{j,h}(h_1) v(h_2)}{1 + 2\eta z_j u_{j,h}(h_1)}.$$

Use transition density one and fair Rademacher outcomes off regime, and on $A_1 = A_2 = 1$ let $P(Y = 1 \mid H_2) = \{1 + \delta v(H_2)\}/2$. Generate the full-data extension from independent uniforms $U_{A_1}, U_{H_2}, U_{A_2}, U_Y^{a_1 a_2}$ by thresholding the treatment probabilities, inverse-transforming each conditional transition, defining the potential outcomes from the corresponding conditional means, and imposing $H_2 = H_2(A_1)$ and $Y = Y(A_1, A_2)$. Let ν_+ mix uniformly over z with $\lambda = z$, let ν_- mix uniformly with $\lambda = -z$, and define $\overline{P}_\pm = \int P_{z,\pm z}^{\otimes n} d\nu_\pm(z)$.

For the same-subclass comparison, fix

$$q(h_2) = \mathbf{1}\{h_{2,1} \leq 1/2\} - \mathbf{1}\{h_{2,1} > 1/2\}, \quad a(h_1) = \cos(2\pi h_{1,1}), \quad v(h_2) = \cos(4\pi h_{2,1}),$$

extended constantly in the remaining coordinates, and fix small $\kappa, \delta > 0$. Define

$$g_0(h_2 \mid h_1) = 1 + \kappa a(h_1) q(h_2).$$

With the same fixed off-regime and independent-uniform full-data laws, define the n -independent subclass

$$\mathcal{M}_{\text{rough}} = \left\{ P \in \mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c) : \begin{array}{l} H_1 \sim \text{Unif}([0, 1]^d), \pi_2 = 1/2, m_2 = \delta v, \\ g = g_0 + r, \int r(h_2 | h_1) dh_2 = 0, h_2 \mapsto r(h_2 | h_1) \text{ is continuous,} \\ (1 - \kappa)/2 \leq g(h_2 | h_1) \leq 2(1 + \kappa) \end{array} \right\}$$

The rough fuzzy transition is

$$g_{z, \lambda}^{\text{rough}}(h_2 | h_1) = g_0(h_2 | h_1) + \epsilon \sum_j \frac{\lambda_j u_{j, h}(h_1) v(h_2)}{1 + 2\eta z_j u_{j, h}(h_1)}.$$

It uses the same $\pi_{1, z}$, π_2 , m_2 , full-data construction, and mixtures $\lambda = z$ versus $\lambda = -z$. Relative to its common law with first propensity $1/2$ and treated transition g_0 , define the cell channels

$$F_j = 2\eta(2A_1 - 1)u_{j, h}(H_1), \quad E_j = \epsilon \mathbf{1}\{A_1 = 1\} \frac{u_{j, h}(H_1)v(H_2)}{g_0(H_2 | H_1)}.$$

For each $P \in \mathcal{M}_{\text{rough}}$, let

$$K_P(dy | l_0, l_1) = P(Y \in dy | H_1 = l_0, H_2 = l_1, A_1 = A_2 = 1)$$

be a regular conditional probability kernel. Such a kernel exists because the state and outcome spaces are standard Borel; strict overlap makes the treated-regime conditioning well-defined on the $p_1(l_0)g(l_1 | l_0)$ -support, and an arbitrary measurable version may be fixed off that support. Define an auxiliary two-occasion full-data law by first drawing (L_0, L_1) with density $p_1(l_0)g(l_1 | l_0)$, then drawing Y^{MAR} from $K_P(\cdot | L_0, L_1)$. Conditional on $(L_0, L_1, Y^{\text{MAR}})$, draw R_0 as Bernoulli with success probability $\pi_1(L_0)$. Set $R_1 = 0$ whenever $R_0 = 0$; conditional on $R_0 = 1$, draw R_1 as Bernoulli with success probability $\pi_2(L_1)$, independently of Y^{MAR} given (L_0, L_1) . Thus the full-data MAR identities are

$$P(R_0 = 1 | L_0, L_1, Y^{\text{MAR}}) = \pi_1(L_0), \quad P(R_1 = 1 | R_0 = 1, L_0, L_1, Y^{\text{MAR}}) = \pi_2(L_1).$$

The auxiliary observed-data law satisfies

$$(R_0, L_0, R_0 L_1, R_1, R_0 R_1 Y^{\text{MAR}}) \stackrel{d}{=} (A_1, H_1, A_1 H_2, A_1 A_2, A_1 A_2 Y).$$

Accordingly, the identified coordinates are $L_0 = H_1$ and $R_0 = A_1$, together with $L_1 = H_2$ and $R_1 = A_2$ only on $A_1 = 1$, and the outcome kernel agrees with the longitudinal observed Y -kernel only on $A_1 = A_2 = 1$. The auxiliary variable L_1 is not identified with the causal potential state $H_2(1)$, and the construction imposes no additional state-consistency condition on the broad causal model. (inputs: $\backslash(d\backslash)$, $\backslash(\backslash\alpha.1\backslash)$, $\backslash(t\backslash)$, $\backslash(M\backslash)$, $\backslash(c\backslash)$)

Definition 13 (def:second-fuzzy-experiment). *period-two paired fuzzy experiment* := Fix $p_1 = 1$, $\pi_1 = 1/2$, and every transition density at one. Choose disjoint

smooth localized bumps $v_{j,h}$ whose reference bump is sign-symmetric under a measure-preserving reflection, so $v_{j,h} \circ \tau_j = -v_{j,h}$. Set $J \asymp h^{-d}$, $a = c_a h^{\alpha_2}$, $b_0 = c_b h^\beta$,

$$\pi_{2,z} = 1/2 + a \sum_j z_j v_{j,h}, \quad m_{2,z,\lambda} = b_0 \sum_j \frac{\lambda_j v_{j,h}}{1 + 2a z_j v_{j,h}}.$$

Use the factorization, off-regime laws, independent-uniform full-data extension, and product mixtures specified in `def:first-fuzzy-experiment`, replacing the terminal treated mean by $m_{2,z,\lambda}$. For $2a\|v_{j,h}\|_\infty < 1$ and $b_0\|v_{j,h}\|_\infty/(1 - 2a\|v_{j,h}\|_\infty) < 1$, all probabilities are valid. Reflection symmetry gives

$$\int \frac{z_j v_{j,h}}{1 + 2a z_j v_{j,h}} = -2a \int \frac{v_{j,h}^2}{1 - 4a^2 v_{j,h}^2}.$$

Thus, with $h \asymp n^{-2/\{2(\alpha_2+\beta)+d\}}$, the two mixing supports have the exact sign-independent separation

$$4ab_0 \sum_j \int \frac{v_{j,h}^2}{1 - 4a^2 v_{j,h}^2} \asymp ab_0 = n^{-r_2},$$

and the affinity scale is $n^2 a^2 b_0^2 / J = O(1)$. (inputs: $\backslash(d\backslash)$, $\backslash(\backslash\alpha_2\backslash)$, $\backslash(\backslash\beta\backslash)$, $\backslash(M\backslash)$, $\backslash(c\backslash)$)

Definition 14 (def:analytic-outgoing-subclass). $\mathcal{G}_{\text{AH}}^{t_0}(M, \theta_0) := v_0(h_2) = \sqrt{2} \cos(2\pi h_{2,1})$ and $\mathcal{G}_{\text{AH}}^{t_0}(M, \theta_0) = \{1 + \theta w(h_1) v_0(h_2) : \|w\|_{C^{t_0}} \leq M, |\theta| \leq \theta_0\}$, where $t_0 > d/2$ is finite and $\theta_0 M \|v_0\|_\infty < 1$. For noninteger t_0 , take $w(h_1) = \chi(h_1) |h_{1,1} - 1/2|^{t_0}$, rescaled to the ball; it is not in $C^{t_0+\xi}$ for any specified $\xi > 0$. (inputs: $\backslash(t_0\backslash)$, $\backslash(M\backslash)$, $\backslash(\backslash\theta\theta_0\backslash)$, $\backslash(w\backslash)$, $\backslash(v_0\backslash)$, $\backslash(\backslash\chi\backslash)$, $\backslash(\backslash x\backslash)$)

6 Main results

Lemma 1 (lem:gformula). For every $P \in \mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c)$, $\psi(P) = E_{Q_1}[m_1(H_1)] = E_{Q_2}[m_2(H_2)]$.

Proof. Fix $P \in \mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c)$, and fix a full-data extension $\tilde{P}$ supplied by `def:longitudinal-model`. Strict positivity makes every treated conditional expectation below well defined. First-period exchangeability and the tower property give

$$E_{\tilde{P}}[Y(1, 1)] = E_{P_{H_1}}[E_{\tilde{P}}\{Y(1, 1) \mid H_1, A_1 = 1\}].$$

Conditioning further on H_2 , using its conditional density g among units with $A_1 = 1$, and applying second-period exchangeability give, for P_{H_1} -almost every h_1 ,

$$E_{\tilde{P}}\{Y(1, 1) \mid H_1 = h_1, A_1 = 1\} = \int E_{\tilde{P}}\{Y(1, 1) \mid H_1 = h_1, H_2 = h_2, A_1 = 1, A_2 = 1\} g(h_2 \mid h_1) dh_2.$$

On $\{A_1 = A_2 = 1\}$, consistency gives $Y = Y(1, 1)$. Outcome-state sufficiency therefore identifies the conditional expectation in the integrand with $m_2(h_2)$. Consequently,

$$E_{\bar{P}}[Y(1, 1)] = E_{Q_1} \left[\int m_2(h_2) g(h_2 \mid H_1) dh_2 \right] = E_{Q_1}[m_1(H_1)].$$

Because $|m_2| \leq 1$, Fubini's theorem applies. The definition of Q_2 then yields

$$E_{Q_1}[m_1(H_1)] = \int m_2(h_2) \left\{ \int g(h_2 \mid h_1) Q_1(dh_1) \right\} dh_2 = E_{Q_2}[m_2(H_2)].$$

This proves both asserted equalities. $\square$

Justification. This identifies the causal target with the observed longitudinal g-formula before any estimator or rate analysis. *Gap.* Identification is standard and is not the statistical contribution. *Consumer.* All upper- and lower-bound calculations evaluate the same observed-data functional.

Lemma 2 (lem:direct-bias). *For $O_1, \dots, O_n \stackrel{\text{iid}}{\sim} P$ with $P \in \mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c)$, conditionally on $\mathcal{F}_{TB}$, $b_1^\dagger = m_1 + T_g\{(1 - w_2)(\hat{m}_2 - m_2)\}$ and*

$$\begin{aligned} E_P[\hat{\psi}^{\text{SDR}} - \psi \mid \mathcal{F}_{TB}] &= E_{Q_1}[(1 - w_1)(\hat{m}_1 - b_1^\dagger)] + E_{Q_2}[(1 - w_2)(\hat{m}_2 - m_2)] \\ &= E_{Q_1}[(1 - w_1)(\hat{m}_1 - m_1)] + E_{Q_1}[w_1 T_g\{(1 - w_2)(\hat{m}_2 - m_2)\}]. \end{aligned}$$

Equivalently, the bias is $B_1 + B_2 + B_T$, where $B_1 = E_{Q_1}[(1 - w_1)(\hat{m}_1 - m_1)]$, $B_2 = E_{Q_2}[(1 - w_2)(\hat{m}_2 - m_2)]$, and $B_T = -E_{Q_1}[(1 - w_1)T_g\{(1 - w_2)(\hat{m}_2 - m_2)\}]$.

Proof. Condition on $\mathcal{F}_T$, and define $\delta_2 = (1 - w_2)(\hat{m}_2 - m_2)$. Treatment-state sufficiency, outcome-state sufficiency, and the definition of $\hat{D}_2$ imply

$$\begin{aligned} E_P[\hat{D}_2 \mid H_1 = h_1, H_2 = h_2, A_1 = 1, \mathcal{F}_T] &= \hat{m}_2(h_2) + \frac{\pi_2(h_2)}{\hat{\pi}_2(h_2)} \{m_2(h_2) - \hat{m}_2(h_2)\} \\ &= m_2(h_2) + \delta_2(h_2). \end{aligned}$$

Integrating against $g(\cdot \mid h_1)$ gives

$$b_1^\dagger(h_1) = m_1(h_1) + T_g \delta_2(h_1).$$

The evaluation fold is independent of $\mathcal{F}_{TB}$. Therefore its empirical average has, conditionally on $\mathcal{F}_{TB}$, the expectation of one fresh observation. Iterated expectation and $E_P[A_1 \mid H_1] = \pi_1(H_1)$ give

$$\begin{aligned} E_P[\hat{\psi}^{\text{SDR}} \mid \mathcal{F}_{TB}] &= E_{Q_1}[\hat{m}_1 + w_1(b_1^\dagger - \hat{m}_1)] \\ &= E_{Q_1}[b_1^\dagger + (1 - w_1)(\hat{m}_1 - b_1^\dagger)]. \end{aligned}$$

By the definition of Q_2 and Fubini's theorem,

$$E_{Q_1}[T_g \delta_2] = E_{Q_2}[\delta_2].$$

Subtracting $\psi = E_{Q_1}[m_1]$, proved in `lem:gformula`, yields

$$E_P[\hat{\psi}^{\text{SDR}} - \psi \mid \mathcal{F}_{TB}] = E_{Q_1}[(1 - w_1)(\hat{m}_1 - b_1^\dagger)] + E_{Q_2}[(1 - w_2)(\hat{m}_2 - m_2)].$$

Substituting $b_1^\dagger = m_1 + T_g \delta_2$ and again using $E_{Q_1}[T_g \delta_2] = E_{Q_2}[\delta_2]$ gives

$$\begin{aligned} E_P[\hat{\psi}^{\text{SDR}} - \psi \mid \mathcal{F}_{TB}] &= E_{Q_1}[(1 - w_1)(\hat{m}_1 - m_1)] + E_{Q_1}[w_1 T_g \delta_2] \\ &= B_1 + B_2 + B_T, \end{aligned}$$

because $B_2 = E_{Q_1}[T_g \delta_2]$ and $B_T = -E_{Q_1}[(1 - w_1)T_g \delta_2]$. This proves every displayed identity. $\square$

Justification. The exact conditional identity isolates two local bilinear biases and the transported mixed term created by regressing an estimated pseudo-outcome. *Gap.* It is an algebraic first-order decomposition, not a proof that the frozen higher-order corrections attain the converse. *Consumer.* It drives the computable first-order risk bound and identifies the obstacle for a matched construction.

Proposition 1 (`prop:finite-transition-separation`). *Assume $M > 1$, $0 < \theta_0$, and $\theta_0 M \|v_0\|_\infty < 1$, and let finite $t_0 > d/2$. Define*

$$\mathcal{A}_{\text{AH}}^{t_0} = \mathcal{G}_{\text{AH}}^{t_0}(M, \theta_0) \cap \mathcal{G}_d^{t_0}(M), \quad \mathcal{M}_{\text{AH}} = \{P \in \mathcal{M}(\alpha_1, \alpha_2, \beta, t_0; M, c) : g \in \mathcal{A}_{\text{AH}}^{t_0}\}.$$

The class $\mathcal{A}_{\text{AH}}^{t_0}$ contains state-dependent transition densities and has infinite-dimensional affine hull. If t_0 is noninteger, it contains a transition of exact incoming Holder order t_0 . Moreover, for every $\alpha_1 > 0$, $\alpha_1 + t_0 > d/2$; hence whenever $\alpha_2 + \beta \geq d/2$, both channels satisfy the necessary lower-channel elbow inequalities. Consequently, as a conditional corollary only, any future theorem establishing the conjectured matched two-channel rate on $\mathcal{M}_{\text{AH}}$ would imply an $O(n^{-1/2})$ minimax upper rate there. No such upper bound is assumed in the definition of $\mathcal{M}_{\text{AH}}$ or proved by this proposition.

Proof. Let $V_1 = \|v_0\|_{L^1([0,1]^d)}$, fix $\theta_\star = \theta_0/2$, and choose $\varepsilon > 0$ so small that

$$\varepsilon \leq M, \quad 1 + \theta_\star V_1 \varepsilon < M.$$

For every $w \in C^{t_0}([0,1]^d)$ with $\|w\|_{C^{t_0}} \leq \varepsilon$, put

$$g_w(h_2 \mid h_1) = 1 + \theta_\star w(h_1) v_0(h_2).$$

Since $\int v_0(h_2) dh_2 = 0$, each $g_w(\cdot \mid h_1)$ integrates to one. Moreover,

$$g_w(h_2 \mid h_1) \geq 1 - \theta_0 M \|v_0\|_\infty > 0$$

and

$$\|h_1 \mapsto g_w(\cdot | h_1)\|_{C^{t_0}(L^1)} \leq 1 + \theta_* V_1 \|w\|_{C^{t_0}} < M.$$

Thus every such g_w belongs to $\mathcal{A}_{\text{AH}}^{t_0}$. A nonconstant w gives a state-dependent transition. A neighborhood of zero in $C^{t_0}([0, 1]^d)$ contains infinitely many linearly independent functions, and the differences $g_w - g_0 = \theta_* w v_0$ preserve their linear independence. Hence the affine hull is infinite-dimensional.

Suppose $t_0 = k + \gamma$ is noninteger, with $k \in \mathbb{N}_0$ and $0 < \gamma < 1$. Choose the cutoff from `def:analytic-outgoing-subclass` and rescale

$$w_*(h_1) = \chi(h_1) |h_{1,1} - 1/2|^{t_0}$$

so that $0 < \|w_*\|_{C^{t_0}} \leq \varepsilon$. Its k -th derivative in the first coordinate has Hölder exponent exactly γ at $h_{1,1} = 1/2$; hence $w_* \in C^{t_0}$ but $w_* \notin C^{t_0+\xi}$ for every $\xi > 0$. The bounded linear functional $L(f) = \int f(h_2) v_0(h_2) dh_2$ on $L^1(dh_2)$ satisfies

$$L\{g_{w_*}(\cdot | h_1) - 1\} = \theta_* w_*(h_1) \int v_0^2 = \theta_* w_*(h_1).$$

If $h_1 \mapsto g_{w_*}(\cdot | h_1)$ belonged to $C^{t_0+\xi}(L^1)$, applying L would force $w_* \in C^{t_0+\xi}$, a contradiction. This transition has exact incoming Hölder order t_0 .

Finally, $t_0 > d/2$ and $\alpha_1 > 0$ imply $s_1 = \alpha_1 + t_0 > d/2$. If $\alpha_2 + \beta \geq d/2$, then $s_2 \geq d/2$. Since

$$\frac{2s}{2s+d} \geq \frac{1}{2} \iff s \geq \frac{d}{2},$$

a future theorem giving the conjectured upper bound $C\{n^{-1/2} + n^{-r_1} + n^{-r_2}\}$ on $\mathcal{M}_{\text{AH}}$ would give at most $3Cn^{-1/2}$. This is only the conditional implication stated in the proposition; the geometric argument supplies no statistical upper bound. $\square$

Justification. This supplies a finite-smoothness, state-dependent transition class with infinite-dimensional affine hull. *Gap.* Its root- n implication remains conditional on a future matched upper theorem. *Consumer.* It illustrates the structural role of incoming transition smoothness without asserting present root- n estimability.

Proposition 2 (prop:single-time-reduction). *If $g(h_2 | h_1) = g_0(h_2)$, then m_1 is constant and $\psi = \int m_2(h_2) g_0(h_2) dh_2$, so the target has no nonconstant incoming-state component. This structural conclusion does not assert that the first-stage terms of the frozen finite-sample estimator vanish identically.*

Proof. If $g(h_2 | h_1) = g_0(h_2)$, then the backward-regression definition gives, for every h_1 ,

$$m_1(h_1) = \int m_2(h_2) g_0(h_2) dh_2.$$

The right-hand side is independent of h_1 , so m_1 is constant. Because Q_1 is a probability measure, the g-formula functional reduces to

$$\psi = E_{Q_1}[m_1(H_1)] = \int m_2(h_2) g_0(h_2) dh_2.$$

This is an identity for the target functional alone. It imposes no identity on fitted propensities, fitted regressions, empirical kernels, or realized corrections, and therefore does not imply that the frozen estimator's first-stage terms vanish in a finite sample. $\square$

Justification. This verifies that a state-independent transition removes the non-constant incoming-state component of the target. *Gap.* It is a target-level reduction and does not simplify every realized estimator term. *Consumer.* It separates the genuinely longitudinal incoming-transition channel from the terminal channel.

Theorem 1 (thm:scalar-correction-order-staircase). *Let $\gamma_j = \alpha_j/(2\alpha_j + d)$, $\gamma_b = \beta/(2\beta + d)$, and $r_\star = \min\{r_1, r_2\}$. The scalar correction-order envelope has the inclusion-minimal finite staircase*

$$\mathcal{I}_\star = \{(p, r) \in \mathbb{N}_0^2 : (p+1)\gamma_1 + (r+1)\gamma_2 + \gamma_b \leq r_\star\}.$$

A rectangle $\{0, \dots, P\} \times \{0, \dots, R\}$ makes the scalar omitted sum $o(n^{-r_\star})$ if and only if

$$(P+2)\gamma_1 + \gamma_2 + \gamma_b > r_\star, \quad \gamma_1 + (R+2)\gamma_2 + \gamma_b > r_\star.$$

No one finite rectangle works uniformly over the full positive asymmetric smoothness orthant. These are exact statements about the deterministic monomial envelope only. The result is an internal deterministic construction diagnostic; it makes no assertion about stochastic higher-order chains, a minimax upper bound, or any published estimator.

Proof. The formula for $\mathcal{I}_\star$, its finiteness and inclusion-minimality, and the necessary-and-sufficient rectangle inequalities follow from **lem:geometric-transport-index-set**. It remains to prove that no single finite rectangle satisfies those inequalities uniformly over the full positive asymmetric smoothness orthant.

Put

$$\Gamma(x) = \frac{x}{2x+d}, \quad \rho(x) = \frac{2x}{2x+d} = 2\Gamma(x), \quad x > 0.$$

Then $\gamma_j = \Gamma(\alpha_j)$, $\gamma_b = \Gamma(\beta)$, and, by the definitions of s_j and r_j ,

$$r_1 = \rho(\alpha_1 + t), \quad r_2 = \rho(\alpha_2 + \beta).$$

The function Γ is continuous and strictly increasing on $(0, \infty)$, since

$$\Gamma'(x) = \frac{d}{(2x+d)^2} > 0.$$

Fix arbitrary finite $P, R \in \mathbb{N}_0$ and any $\varepsilon > 0$. To make the first rectangle inequality fail, set

$$t = \varepsilon, \quad \alpha_2 = \beta = \frac{3\varepsilon}{4},$$

and let $\alpha_1 \downarrow 0$. For all sufficiently small positive α_1 ,

$$s_1 = \alpha_1 + \varepsilon < \frac{3\varepsilon}{2} = s_2,$$

so $r_\star = r_1 = \rho(\alpha_1 + \varepsilon)$. By continuity and strict monotonicity of Γ ,

$$\lim_{\alpha_1 \downarrow 0} \{(P+2)\gamma_1 + \gamma_2 + \gamma_b\} = 2\Gamma(3\varepsilon/4) < 2\Gamma(\varepsilon) = \lim_{\alpha_1 \downarrow 0} r_\star.$$

The strict inequality persists for all sufficiently small positive α_1 . Hence the first necessary rectangle inequality fails at such a point of the positive smoothness orthant.

For completeness, the other rectangle inequality can independently be made to fail. Set

$$\beta = \varepsilon, \quad t = \frac{3\varepsilon}{2}, \quad \alpha_1 = \alpha_2 = u,$$

and let $u \downarrow 0$. Then

$$s_2 = u + \varepsilon < u + \frac{3\varepsilon}{2} = s_1,$$

so $r_\star = r_2 = \rho(u + \varepsilon)$, whereas

$$\lim_{u \downarrow 0} \{\gamma_1 + (R+2)\gamma_2 + \gamma_b\} = \Gamma(\varepsilon) < 2\Gamma(\varepsilon) = \lim_{u \downarrow 0} r_\star.$$

Thus the second necessary rectangle inequality also fails for all sufficiently small positive u . Since P and R were arbitrary, no fixed finite rectangle works uniformly over the full positive asymmetric smoothness orthant.

The argument concerns only the deterministic nonnegative monomial series

$$\sum_{(p,r) \notin \mathcal{I}} a_1^{p+1} a_2^{r+1} b.$$

It does not bound random projection-kernel norms, degenerate U-statistic variances, or stochastic transport remainders. In particular, `lem:stage-two-empirical-gram-rank` shows that a stage-two block whose dimension exceeds the treated Gram-fold count is singular deterministically, and that proportional blocks exceeding the treatment fraction are singular with probability tending to one under a constant first propensity. Therefore the staircase is exactly the deterministic construction diagnostic claimed in the theorem and entails no stochastic higher-order-chain or minimax assertion. $\square$

Justification. The theorem exactly characterizes deterministic monomial correction-index bookkeeping for the frozen construction. *Gap.* It is an internal deterministic diagnostic and proves no stochastic higher-order or minimax claim. *Consumer.* It records constraints relevant to any future analysis of the note's own higher-order chains.

Lemma 3 (`lem:bounded-design-wavelet-regression`). *Let $\mathcal{A}$ be a sigma-field. Conditionally on $\mathcal{A}$, let $(X_i, S_i, W_i)_{i=1}^m$ be independent and identically distributed, with $X_i \in [0, 1]^d$, $S_i \in \{0, 1\}$, $|W_i| \leq L$, and*

$$E[W_i \mid X_i = x, S_i = 1, \mathcal{A}] = f_{\mathcal{A}}(x).$$

Let Q have a Lebesgue density bounded above and away from zero. Suppose, almost surely, that the selected subprobability measure $B \mapsto E[S_i \mathbf{1}\{X_i \in B\} \mid \mathcal{A}]$ has a density whose ratio to the Q -density is bounded above and away from zero by deterministic constants, and that $f_{\mathcal{A}} \in \mathcal{H}_d^s(R)$. Let $\hat{f}$ be the clipped, minimum-norm boundary-corrected wavelet least-squares fit among observations with $S_i = 1$, from a basis with regularity exceeding s , using $k \asymp m^{d/(2s+d)}$ basis functions and a fixed clipping interval containing the range of every $f_{\mathcal{A}}$. Then, almost surely,

$$\left\{ E \left[\|\hat{f} - f_{\mathcal{A}}\|_{L^2(Q)}^4 \mid \mathcal{A} \right] \right\}^{1/4} \leq C \left\{ k^{-s/d} + \sqrt{k/m} \right\} \lesssim m^{-s/(2s+d)}.$$

The constant is uniform over all conditional targets and conditional laws satisfying the displayed bounds.

Proof. Condition throughout on $\mathcal{A}$; all constants below are deterministic because every conditional radius, density ratio, and range bound is uniform. Let V_k be the span of the first k boundary-corrected wavelets and write Z_k for their basis vector. Boundary-wavelet coefficient decay for $f_{\mathcal{A}} \in C^s$, obtained by subtracting on each wavelet support the Taylor polynomial annihilated by the wavelet moments, gives a deterministic truncation $f_k^0 \in V_k$ satisfying

$$\|f_{\mathcal{A}} - f_k^0\|_{L^2(Q)} \leq Ck^{-s/d}, \quad \|f_{\mathcal{A}} - f_k^0\|_{\infty} \leq Ck^{-s/d}, \quad \|f_k^0\|_{\infty} \leq C.$$

These inequalities hold uniformly for every realization of $\mathcal{A}$.

Write $f_k^0(x) = Z_k(x)^T \theta_0$,

$$G = E[SZ_k(X)Z_k(X)^T \mid \mathcal{A}], \quad \hat{G} = \frac{1}{m} \sum_{i=1}^m S_i Z_k(X_i) Z_k(X_i)^T.$$

Lebesgue orthonormality and the conditional selected-density bounds put all eigenvalues of G in a fixed interval $[c_0, C_0]$. Bounded overlap of boundary wavelets gives $\sup_x \|Z_k(x)\|_2^2 \leq Ck$. For $X_i = S_i Z_k(X_i) Z_k(X_i)^T - G$, one has

$$\|X_i\|_{\text{op}} \leq Ck, \quad \left\| \sum_{i=1}^m E[X_i^2 \mid \mathcal{A}] \right\|_{\text{op}} \leq Cmk.$$

Indeed, $\{SZZ^T\}^2 = S\|Z\|_2^2 ZZ^T \preceq Ck SZZ^T$. The matrix exponential inequality

$$e^{\theta X} \preceq I + \theta X + \frac{\theta^2 X^2}{2(1 - \theta Ck/3)}, \quad 0 < \theta < 3/(Ck),$$

followed by the trace-Laplace iteration for the independent centered matrices and optimization over θ , yields

$$P\left(\|\hat{G} - G\|_{\text{op}} > c_0/2 \mid \mathcal{A}\right) \leq 2k \exp(-cm/k).$$

Call the complementary event $\mathcal{E}_k$. On $\mathcal{E}_k$, $\|\hat{G}^{-1}\|_{\text{op}} \leq 2/c_0$ and $\|\hat{G}\|_{\text{op}} \leq C$.

Set

$$\xi_i = S_i Z_k(X_i) \{W_i - f_{\mathcal{A}}(X_i)\}, \quad a_i = f_{\mathcal{A}}(X_i) - f_k^0(X_i).$$

Then $E[\xi_i \mid \mathcal{A}] = 0$, $\|\xi_i\|_2^2 \leq Ck$, $E[\|\xi_i\|_2^2 \mid \mathcal{A}] \leq Ck$, and $E[\|\xi_i\|_2^4 \mid \mathcal{A}] \leq Ck^2$. Expanding the fourth power of the norm of a sum of independent centered vectors, all terms with an index appearing only once vanish, so

$$E \left[\left\| \frac{1}{m} \sum_{i=1}^m \xi_i \right\|_2^4 \mid \mathcal{A} \right] \leq \frac{C}{m^4} \left\{ \left(\sum_i E[\|\xi_i\|_2^2 \mid \mathcal{A}] \right)^2 + \sum_i E[\|\xi_i\|_2^4 \mid \mathcal{A}] \right\} \leq C \frac{k^2}{m^2}.$$

On $\mathcal{E}_k$, expansion of the normal equations around the deterministic approximant gives

$$\hat{\theta} - \theta_0 = \hat{G}^{-1} \left\{ \frac{1}{m} \sum_i \xi_i + \frac{1}{m} \sum_i S_i Z_k(X_i) a_i \right\}.$$

For every unit vector u , empirical Cauchy–Schwarz gives

$$\left| u^T \frac{1}{m} \sum_i S_i Z_k(X_i) a_i \right| \leq \{u^T \hat{G} u\}^{1/2} \left\{ \frac{1}{m} \sum_i S_i a_i^2 \right\}^{1/2} \leq Ck^{-s/d}.$$

Thus norm equivalence between coefficients in V_k and $L^2(Q)$, the fourth-moment bound, and the approximation inequalities imply

$$\left\{ E \left[\|\hat{f} - f_{\mathcal{A}}\|_{L^2(Q)}^4 \mathbf{1}_{\mathcal{E}_k} \mid \mathcal{A} \right] \right\}^{1/4} \leq C \{ \sqrt{k/m} + k^{-s/d} \}.$$

Clipping contracts pointwise distance toward $f_{\mathcal{A}}$. On $\mathcal{E}_k^c$, both the clipped fit and the target are uniformly bounded, so its fourth-moment contribution is at most $C\{2k \exp(-cm/k)\}^{1/4}$. For $k \asymp m^{d/(2s+d)}$, this is smaller than every polynomial in m^{-1} , while $k^{-s/d} \asymp \sqrt{k/m} \asymp m^{-s/(2s+d)}$. Combining the good- and bad-event bounds proves the conditional claim. Taking $\mathcal{A}$ trivial gives the unconditional special case. $\square$

Lemma 4 (lem:treated-design-and-backward-regularity). *For every $P \in \mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c)$, the laws of $H_1 \mid A_1 = 1$ and Q_1 have uniformly bounded density ratios, and the laws of $H_2 \mid A_1 = 1$, $H_2 \mid A_1 = A_2 = 1$, and Q_2 have uniformly bounded density ratios. If $\hat{\pi}_2 \in [c/2, 1 - c/2]$ and $\hat{m}_2 \in [-1, 1]$, then*

$$b_1^\dagger(h_1) = \int \left[\hat{m}_2(h_2) + \frac{\pi_2(h_2)}{\hat{\pi}_2(h_2)} \{m_2(h_2) - \hat{m}_2(h_2)\} \right] g(h_2 \mid h_1) dh_2$$

is bounded by $L_c = 1 + 4/c$ and has a $C^t([0, 1]^d)$ norm bounded by a constant depending only on M and c .

Proof. Let q_j be the densities of Q_j . Since

$$P(H_1 \in dh_1 \mid A_1 = 1) = \frac{q_1(h_1)\pi_1(h_1)}{P(A_1 = 1)} dh_1$$

and both numerator multiplier and denominator lie in $[c, 1 - c]$, this design is uniformly equivalent to Q_1 . Next define

$$\tilde{q}_2(h_2) = \int q_1(h_1) \pi_1(h_1) g(h_2 | h_1) dh_1.$$

Overlap gives $cq_2 \leq \tilde{q}_2 \leq (1 - c)q_2$. Normalizing by $P(A_1 = 1)$ proves equivalence of $H_2 | A_1 = 1$ and Q_2 . Multiplication by $\pi_2(h_2)$ and normalization prove the same for $H_2 | A_1 = A_2 = 1$. Combined with the design-density bounds, every one of these densities is bounded above and away from zero.

For the regularity claim, put

$$F(h_2) = \hat{m}_2(h_2) + \frac{\pi_2(h_2)}{\hat{\pi}_2(h_2)} \{m_2(h_2) - \hat{m}_2(h_2)\}.$$

The clipping and overlap bounds give $\|F\|_\infty \leq 1 + (2/c)2 = L_c$. For any derivative or Hölder difference operator D in the incoming argument of order at most t , Fubini's theorem and the transition-ball condition yield

$$|Db_1^\dagger(h_1)| = \left| \int F(h_2) Dg(h_2 | h_1) dh_2 \right| \leq \|F\|_\infty \|Dg(\cdot | h_1)\|_{L^1}.$$

Taking the supremum over all terms in the C^t norm gives $\|b_1^\dagger\|_{C^t} \leq L_c M$, as required. $\square$

Lemma 5 (lem:symmetric-cell-mixture-affinity). *Let R_0 be a probability law and let bounded functions f, g satisfy $E_0 f = E_0 g = E_0(fg) = 0$ and $\|f\|_\infty + \|g\|_\infty \leq \kappa < 1$. For k observations, mix the product densities $\prod_{i=1}^k (1 + zf_i + \lambda g_i)$ uniformly over $z \in \{-1, 1\}$, with $\lambda = z$ for Q_k^+ and $\lambda = -z$ for Q_k^- . If $A = E_0 f^2$ and $B = E_0 g^2$, then*

$$H^2(Q_k^+, Q_k^-) \leq C_\kappa \min\{kA, kB, k(k-1)AB, 1\}.$$

For J disjoint cells of probabilities p_j , independent signs, and conditional channel norms A_j, B_j , the fixed-sample product mixtures obey

$$H^2(\bar{P}_+, \bar{P}_-) \leq C_\kappa n(n-1) \sum_{j=1}^J p_j^2 A_j B_j.$$

Proof. Write $r_{z,\lambda} = 1 + zf + \lambda g$. Pairing $r_{z,z}$ with $r_{z,-z}$, convexity of squared Hellinger distance and product expansion give $H^2(Q_k^+, Q_k^-) \leq C_\kappa kB$. Reindexing the minus mixture by $z \mapsto -z$ and pairing $r_{z,z}$ with $r_{-z,z}$ gives the symmetric bound $C_\kappa kA$; the bound by one is automatic.

For the product bound, give latent signs Z, L the correlation law $P_\gamma(Z = z, L = \ell) = (1 + \gamma z \ell)/4$. The observed mixture density is affine in γ , say $q_\gamma = q_0 + \gamma q_1$, and $|q_1| \leq q_0$. Hence

$$H^2(Q_k^+, Q_k^-) \leq 2 \int q_1^2 / q_0.$$

Expanding two likelihood replicas, the coefficient ZL cancels every term containing only one channel, and $E_0(fg) = 0$ cancels a mixed pair on the same observation. Every surviving monomial therefore uses an f -factor and a g -factor on distinct observations. When kA and kB are bounded, Cauchy–Schwarz and the binomial expansion give $\int q_1^2/q_0 \leq C_\kappa k(k-1)AB$. If kA is large, the bound $C_\kappa kB$ is already bounded by a constant multiple of $k(k-1)AB$; the case of large kB is symmetric.

Condition on multinomial cell counts $N_1, \dots, N_J$. Independent signs make the conditional mixtures tensor products across cells, so the preceding bound gives

$$H^2(\bar{P}_+, \bar{P}_- \mid N_1, \dots, N_J) \leq C_\kappa \sum_j N_j(N_j - 1)A_jB_j.$$

The count law is common to both mixtures and $E[N_j(N_j - 1)] = n(n-1)p_j^2$. Averaging proves the last display. $\square$

Lemma 6 (lem:robins-theorem-forty-nine-smooth-density-condition). *In Section 7.2.2 of Robins, Li, Tchetgen Tchetgen, and van der Vaart [3], Theorem 49 is stated under conditions (a1)–(c2). Condition (c2) requires the marginal density of (L_0, L_1) to belong to a positive-order Holder ball. Consequently, a monotone-MAR law whose joint (L_0, L_1) -density has a genuine jump is not covered by the stated assumptions of Theorem 49.*

Justification. This cited leaf records the exact public smooth-joint-density condition used by the comparator proposition. *Gap.* It is cited substrate, not a theorem proved in this note. *Consumer.* It supports the precise noncoverage clause for the completed auxiliary monotone-MAR laws.

Proposition 3 (prop:rough-transition-comparator-separation). *Assume $M > 1$ and $0 < \underline{q} < 1 < \bar{q} < \infty$. There are fixed sufficiently small constants $\kappa, \delta, c_\eta, c_\epsilon > 0$ for which the class $\mathcal{M}_{\text{rough}}$ in `def:first-fuzzy-experiment` is nonempty, independent of n , contained in $\mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c)$, and contains every sufficiently large- n member of the rough period-one fuzzy experiment. For every $P \in \mathcal{M}_{\text{rough}}$, the construction in `def:first-fuzzy-experiment` produces an auxiliary two-occasion monotone-MAR full-data law whose (L_0, L_1) -density is $f_{L_0, L_1}(l_0, l_1) = p_1(l_0)g(l_1 \mid l_0)$, whose conditional outcome law is $K_P(dy \mid l_0, l_1)$, and which satisfies*

$$P(R_0 = 1 \mid L_0, L_1, Y^{\text{MAR}}) = \pi_1(L_0), \quad P(R_1 = 1 \mid R_0 = 1, L_0, L_1, Y^{\text{MAR}}) = \pi_2(L_1),$$

with $R_1 = 0$ whenever $R_0 = 0$. Its observed-data law obeys

$$(R_0, L_0, R_0L_1, R_1, R_0R_1Y^{\text{MAR}}) \stackrel{d}{=} (A_1, H_1, A_1H_2, A_1A_2, A_1A_2Y).$$

Thus $L_0 = H_1$ and $R_0 = A_1$ in the identified coordinates, while $L_1 = H_2$ and $R_1 = A_2$ only on $A_1 = 1$, and the outcome kernel agrees with the longitudinal observed outcome law only on $A_1 = A_2 = 1$. This construction neither identifies L_1 with the causal potential state $H_2(1)$ nor imposes a state-consistency

condition on the broad causal model. Since $p_1 \equiv 1$ and $g = g_0 + r$ throughout $\mathcal{M}_{\text{rough}}$,

$$f_{L_0, L_1} = p_1 g = g_0 + r.$$

This density is bounded and bounded away from zero but has a genuine non-removable outgoing-state jump. Therefore the stated condition (c2) of Robins *et al.* does not cover the induced laws from this subclass. Nevertheless, for all sufficiently large n ,

$$\inf_{\hat{\psi}} \sup_{P \in \mathcal{M}_{\text{rough}}} E_P[(\hat{\psi} - \psi(P))^2]^{1/2} \geq Cn^{-r_1},$$

where $C > 0$ is independent of n . This asserts only noncoverage by the stated assumptions of Theorem 49, not reverse inclusion, priority, or broad non-overlap.

Proof. Put $q(h_2) = \mathbf{1}\{h_{2,1} \leq 1/2\} - \mathbf{1}\{h_{2,1} > 1/2\}$, $a(h_1) = \cos(2\pi h_{1,1})$, and $v(h_2) = \cos(4\pi h_{2,1})$. Direct integration gives $\int q = \int a = \int v = \int qv = 0$. The function q has a genuine jump, whereas a and v are analytic. For $g_0 = 1 + \kappa a q$,

$$1 - \kappa \leq g_0 \leq 1 + \kappa, \quad \|h_1 \mapsto g_0(\cdot | h_1)\|_{C^t(L^1)} \leq 1 + C_t \kappa.$$

Choose fixed $\kappa > 0$ within the slack $M - 1$, and then choose fixed $\delta > 0$ so that $\delta v \in \mathcal{H}_d^\beta(M)$ and $\delta \|v\|_\infty < 1$. With uniform H_1 , fair treatments, $g = g_0$, and $m_2 = \delta v$, the fixed off-regime laws and the independent-uniform full-data extension in `def:first-fuzzy-experiment` satisfy consistency, sequential exchangeability, both state-sufficiency restrictions, and strict overlap. Moreover,

$$\frac{dQ_2}{dh_2} = \int g_0(h_2 | h_1) dh_1 = 1 + \kappa q(h_2) \int a(h_1) dh_1 = 1.$$

Thus all design-density bounds hold, and $\mathcal{M}_{\text{rough}}$ is fixed, nonempty, and contained in $\mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c)$.

We next verify membership of the fuzzy alternatives in this same fixed subclass. Disjoint support, the usual rescaling of a fixed smooth bump, and denominators bounded away from zero give, uniformly in the signs,

$$\left\| \sum_j z_j u_{j,h} \right\|_{C^{\alpha_1}} \leq Ch^{-\alpha_1}, \quad \left\| \sum_j \frac{\lambda_j u_{j,h} v}{1 + 2\eta z_j u_{j,h}} \right\|_{C^t(L^1)} \leq Ch^{-t}.$$

Hence $\eta = c_\eta h^{\alpha_1}$ and $\epsilon = c_\epsilon h^t$, with fixed sufficiently small $c_\eta, c_\epsilon > 0$, preserve the propensity Holder ball, strict overlap, and the transition-ball slack. Normalization follows from $\int q = \int v = 0$, while

$$g_{z,\lambda}^{\text{rough}} \geq 1 - \kappa - \frac{\epsilon \|u\|_\infty \|v\|_\infty}{1 - 2\eta \|u\|_\infty} > 0$$

for all sufficiently large n . The perturbation of g_0 is continuous in h_2 , integrates to zero in h_2 , and, after making the fixed amplitudes smaller if necessary, obeys

the two pointwise bounds in the definition of $\mathcal{M}_{\text{rough}}$. The stated factorization and independent-uniform extension again supply all causal and state-sufficiency conditions. Reflection oddness and $\int u_{j,h} = 0$ imply

$$\left| \int \frac{u_{j,h}(h_1)}{1 + 2\eta z_j u_{j,h}(h_1)} dh_1 \right| \leq C\eta h^d.$$

Because $Jh^d \asymp 1$, it follows that

$$\left\| \frac{dQ_2}{dh_2} - 1 \right\|_{\infty} \leq C\eta\epsilon.$$

The strict slack $\underline{q} < 1 < \bar{q}$ therefore places every sufficiently large- n fuzzy alternative in $\mathcal{M}_{\text{rough}}$.

For any $P \in \mathcal{M}_{\text{rough}}$, use the regular conditional kernel K_P fixed in `def:first-fuzzy-experiment`. Draw (L_0, L_1) with joint density $p_1(l_0)g(l_1 | l_0)$, and then draw Y^{MAR} from $K_P(\cdot | L_0, L_1)$. Conditional on $(L_0, L_1, Y^{\text{MAR}})$, draw R_0 as Bernoulli with success probability $\pi_1(L_0)$. Set $R_1 = 0$ on $\{R_0 = 0\}$; on $\{R_0 = 1\}$, draw R_1 as Bernoulli with success probability $\pi_2(L_1)$, independently of Y^{MAR} given (L_0, L_1) . The construction directly gives

$$P(R_0 = 1 | L_0, L_1, Y^{\text{MAR}}) = \pi_1(L_0)$$

and

$$P(R_1 = 1 | R_0 = 1, L_0, L_1, Y^{\text{MAR}}) = \pi_2(L_1).$$

To verify the coarsened-law assertion, first condition on $R_0 = 1, L_0 = l_0$. Then L_1 has density $g(\cdot | l_0)$, $R_1 | L_1 = l_1$ has success probability $\pi_2(l_1)$, and, on $R_1 = 1$, Y^{MAR} has kernel $K_P(\cdot | l_0, l_1)$. These are respectively the longitudinal conditional laws of H_2 , A_2 , and Y on the observed treated regimes, by the factorization, treatment-state sufficiency, and the definition of K_P . On $R_0 = 0$, the coordinates $R_0 L_1$, R_1 , and $R_0 R_1 Y^{\text{MAR}}$ all equal zero, just as $A_1 H_2$, $A_1 A_2$, and $A_1 A_2 Y$ do on $A_1 = 0$. Since (R_0, L_0) and (A_1, H_1) also have the same law, it follows that

$$(R_0, L_0, R_0 L_1, R_1, R_0 R_1 Y^{\text{MAR}}) \stackrel{d}{=} (A_1, H_1, A_1 H_2, A_1 A_2, A_1 A_2 Y).$$

Thus the identified coordinates equate L_0 with H_1 and R_0 with A_1 , but equate L_1 with H_2 and R_1 with A_2 only on $A_1 = 1$; likewise, K_P agrees with the longitudinal outcome kernel only on $A_1 = A_2 = 1$. In particular, this auxiliary completion neither identifies L_1 with the causal potential state $H_2(1)$ nor imposes any additional state-consistency condition on the broad causal model.

The joint (L_0, L_1) -density in this auxiliary law is $p_1 g = g_0 + r$, because $p_1 \equiv 1$. The function r is continuous in h_2 . Across $h_{2,1} = 1/2$, the right trace minus the left trace of $g_0 + r$ is therefore

$$-2\kappa a(h_1),$$

which is nonzero on open sets of h_1 . No null-set modification can reconcile these one-sided traces. Thus this density is bounded and bounded away from zero but has no continuous representative, and hence belongs to no positive-order Holder ball. The noncoverage conclusion for these completed auxiliary monotone-MAR laws now follows from **lem:robins-theorem-forty-nine-smooth-density-condition**.

It remains to prove the risk floor. Relative to the common law having fair first treatment and treated transition g_0 , the likelihood ratio on cell j is $1 + z_j F_j + \lambda_j E_j$, where

$$F_j = 2\eta(2A_1 - 1)u_{j,h}(H_1), \quad E_j = \epsilon \mathbf{1}\{A_1 = 1\} \frac{u_{j,h}(H_1)v(H_2)}{g_0(H_2 | H_1)}.$$

Indeed, on $A_1 = 1$,

$$\frac{\pi_{1,z} g_{z,\lambda}^{\text{rough}}}{(1/2)g_0} = 1 + 2\eta z_j u_{j,h} + \epsilon \lambda_j u_{j,h} v / g_0,$$

and on $A_1 = 0$ the ratio is $1 - 2\eta z_j u_{j,h}$. Fair treatment, $\int u_{j,h} = 0$, and integration against g_0 give

$$E_0 F_j = E_0 E_j = E_0(F_j E_j) = 0.$$

The fixed lower and upper bounds on g_0 make the conditional squared channel norms comparable to η^2 and ϵ^2 . Since the cell probabilities satisfy $p_j \asymp J^{-1}$, **lem:symmetric-cell-mixture-affinity** yields

$$H^2(\bar{P}_+, \bar{P}_-) \leq C n^2 \eta^2 \epsilon^2 / J.$$

The target under the common base is zero because $\iint \delta v g_0 = \delta \int v + \delta \kappa(\int a)(\int qv) = 0$. Writing $I_v = \int v^2 > 0$, every alternative satisfies

$$\psi_{z,\lambda} = \delta \epsilon I_v \sum_j \lambda_j \int \frac{u_{j,h}}{1 + 2\eta z_j u_{j,h}}.$$

Reflection symmetry gives the exact identity

$$\int \frac{z_j u_{j,h}}{1 + 2\eta z_j u_{j,h}} = -2\eta \int \frac{u_{j,h}^2}{1 - 4\eta^2 u_{j,h}^2}.$$

Consequently, the $\lambda = z$ mixing support has target $-D_h$, the $\lambda = -z$ support has target D_h , and $D_h \asymp \eta \epsilon$, since $J h^d \asymp 1$. Choose

$$h \asymp n^{-2/\{2(\alpha_1+t)+d\}}.$$

Then $n^2 \eta^2 \epsilon^2 / J = O(1)$ and

$$D_h \asymp \eta \epsilon = h^{\alpha_1+t} \asymp n^{-2(\alpha_1+t)/\{2(\alpha_1+t)+d\}} = n^{-r_1}.$$

Taking the fixed amplitude constants sufficiently small makes the affinity $\int \sqrt{d\bar{P}_+ d\bar{P}_-}$ at least some $\rho > 0$, uniformly in n . For any estimator $\hat{\psi}$, the test $\phi = \mathbf{1}\{\hat{\psi} \geq 0\}$ incurs squared error at least D_h^2 on $\{\phi = 1\}$ under $\bar{P}_+$ and on $\{\phi = 0\}$ under $\bar{P}_-$. Hence

$$\frac{1}{2}E_{\bar{P}_+}(\hat{\psi} + D_h)^2 + \frac{1}{2}E_{\bar{P}_-}(\hat{\psi} - D_h)^2 \geq \frac{D_h^2}{2}\{1 - \text{TV}(\bar{P}_+, \bar{P}_-)\}.$$

Cauchy–Schwarz gives $\text{TV}(\bar{P}_+, \bar{P}_-) \leq \sqrt{1 - \rho^2}$. Each mixture is supported on iid laws in the fixed class $\mathcal{M}_{\text{rough}}$, so the left side is bounded above by the worst-case mean squared error over that class. Taking square roots and then the infimum over estimators proves

$$\inf_{\hat{\psi}} \sup_{P \in \mathcal{M}_{\text{rough}}} E_P[(\hat{\psi} - \psi(P))^2]^{1/2} \geq Cn^{-r_1}$$

for a constant $C > 0$ independent of n . $\square$

Justification. This proves the incoming n^{-r_1} floor on a fixed incoming-smooth but outgoing-rough subclass and specifies the comparison on a complete auxiliary monotone-MAR law with outcome kernel K_P . *Gap.* The comparison proves only failure of the stated smooth-joint-density condition; it proves neither reverse inclusion nor a matched upper rate. *Consumer.* It shows that the transition-mediated lower channel is not an artifact of globally smooth outgoing-state densities while keeping the auxiliary and causal full-data laws distinct.

Lemma 7 (lem:stage-two-empirical-gram-rank). *Let $N_1 = \sum_{i \in \mathcal{G}} A_{1i}$. For every stage-two block of dimension k_b and every choice of finite nonzero weights on treated observations,*

$$\text{rank}(\hat{\Omega}_{2,b}) \leq \min\{k_b, N_1\}.$$

Consequently, $k_b > N_1$ implies singularity deterministically. If $m = |\mathcal{G}| \rightarrow \infty$, $\pi_1 \equiv p$, and $k_b/m \rightarrow \lambda > p$, then $P(\hat{\Omega}_{2,b} \text{ is singular}) \rightarrow 1$.

Proof. Let $m = |\mathcal{G}|$. By `def:blocked-projection`, the stage-two empirical Gram matrix is

$$\hat{\Omega}_{2,b} = \frac{1}{m} \sum_{i \in \mathcal{G}} \frac{A_{1i}}{\hat{\pi}_1(H_{1i})} Z_{2,b}(H_{2i}) Z_{2,b}(H_{2i})^T.$$

Each summand with $A_{1i} = 1$ has rank at most one, and each summand with $A_{1i} = 0$ is zero. There are exactly N_1 nonzero candidate summands. Rank subadditivity and the fact that $\hat{\Omega}_{2,b}$ is a $k_b \times k_b$ matrix therefore give

$$\text{rank}(\hat{\Omega}_{2,b}) \leq N_1, \quad \text{rank}(\hat{\Omega}_{2,b}) \leq k_b.$$

The same argument applies when the treated outer products are multiplied by arbitrary finite nonzero scalar weights, because scalar multiplication cannot increase the rank of a rank-one outer product. Hence

$$\text{rank}(\widehat{\Omega}_{2,b}) \leq \min\{k_b, N_1\}.$$

If $k_b > N_1$, this rank is strictly smaller than the matrix dimension k_b , so $\widehat{\Omega}_{2,b}$ is singular.

For the asymptotic assertion, suppose $\pi_1 \equiv p$. By `ass:iid`, the variables $(A_{1i})_{i \in \mathcal{G}}$ are independent Bernoulli random variables with mean p . The weak law of large numbers yields $N_1/m \rightarrow p$ in probability. If $k_b/m \rightarrow \lambda > p$, choose $\varepsilon = (\lambda - p)/3$. With probability tending to one,

$$\frac{N_1}{m} < p + \varepsilon < \lambda - \varepsilon < \frac{k_b}{m}.$$

Thus $N_1 < k_b$ with probability tending to one. The deterministic implication above then gives

$$P\{\widehat{\Omega}_{2,b} \text{ is singular}\} \rightarrow 1.$$

This proves every assertion. $\square$

Justification. The rank inequality is a correct finite-sample linear-algebra diagnostic for the displayed empirical matrix. *Gap.* It says nothing by itself about a rank-stable replacement or another author's estimator. *Consumer.* It is used only to audit internal block choices.

Lemma 8 (lem:geometric-transport-index-set). *Let $\gamma_j = \alpha_j/(2\alpha_j + d)$, $\gamma_b = \beta/(2\beta + d)$, $r_\star = \min\{r_1, r_2\}$, $a_j = n^{-\gamma_j}$, and $b = n^{-\gamma_b}$. Then*

$$\mathcal{I}_\star = \{(p, r) \in \mathbb{N}_0^2 : (p+1)\gamma_1 + (r+1)\gamma_2 + \gamma_b \leq r_\star\}$$

is finite and inclusion-minimal among index sets $\mathcal{I}$ satisfying

$$\sum_{(p,r) \notin \mathcal{I}} a_1^{p+1} a_2^{r+1} b = o(n^{-r_\star}).$$

A rectangle $\{0, \dots, P\} \times \{0, \dots, R\}$ satisfies the condition if and only if

$$(P+2)\gamma_1 + \gamma_2 + \gamma_b > r_\star, \quad \gamma_1 + (R+2)\gamma_2 + \gamma_b > r_\star.$$

Proof. Define

$$E(p, r) = \gamma_b + (p+1)\gamma_1 + (r+1)\gamma_2, \quad (p, r) \in \mathbb{N}_0^2.$$

By `def:preliminary-rates`, $a_j = n^{-\gamma_j}$ and $b = n^{-\gamma_b}$, so

$$a_1^{p+1} a_2^{r+1} b = n^{-E(p,r)}.$$

Because $\alpha_1, \alpha_2, \beta, d > 0$, one has $\gamma_1, \gamma_2, \gamma_b > 0$. Consequently, $E(p, r) \leq r_\star$ bounds both p and r , and hence $\mathcal{I}_\star$ is finite.

The set $\mathcal{I}_\star$ is a coordinatewise lower set, so its complement is a coordinatewise upper set. Let $\mathcal{F}$ be the set of coordinatewise-minimal elements of $\mathbb{N}_0^2 \setminus \mathcal{I}_\star$. This frontier is nonempty and finite. Indeed, every frontier point (p, r) satisfies either $p = 0$, $r = 0$, or both immediate predecessors belong to the finite set $\mathcal{I}_\star$; there is only one minimal complement point on each axis, and the remaining frontier points are therefore bounded coordinatewise by elements of $\mathcal{I}_\star$. Every point outside $\mathcal{I}_\star$ dominates at least one point of $\mathcal{F}$. Moreover, $E(p, r) > r_\star$ for every $(p, r) \in \mathcal{F}$. Since $\mathcal{F}$ is finite,

$$\delta = \min_{(p,r) \in \mathcal{F}} \{E(p, r) - r_\star\} > 0.$$

Covering the complement by the upper-right quadrants generated by $\mathcal{F}$ and summing geometric series gives

$$\begin{aligned} \sum_{(p,r) \notin \mathcal{I}_\star} a_1^{p+1} a_2^{r+1} b &\leq \sum_{(p_0, r_0) \in \mathcal{F}} \sum_{p \geq p_0, r \geq r_0} n^{-E(p, r)} \\ &= \sum_{(p_0, r_0) \in \mathcal{F}} \frac{n^{-E(p_0, r_0)}}{(1 - n^{-\gamma_1})(1 - n^{-\gamma_2})} \\ &\leq \frac{|\mathcal{F}| n^{-r_\star - \delta}}{(1 - n^{-\gamma_1})(1 - n^{-\gamma_2})} = o(n^{-r_\star}). \end{aligned}$$

Thus $\mathcal{I}_\star$ has the required omitted-sum property. Conversely, if an index set $\mathcal{I}$ omits some $(p, r) \in \mathcal{I}_\star$, then its nonnegative complement sum contains $n^{-E(p, r)}$ with $E(p, r) \leq r_\star$. That monomial is not $o(n^{-r_\star})$. Therefore every index set having the omitted-sum property contains $\mathcal{I}_\star$, which proves inclusion-minimality.

Now fix $P, R \in \mathbb{N}_0$ and put $\mathcal{R}_{P, R} = \{0, \dots, P\} \times \{0, \dots, R\}$. Splitting its complement into the part with $p \geq P+1$ and the remaining part with $r \geq R+1$, and then enlarging the latter part, yields

$$\sum_{(p,r) \notin \mathcal{R}_{P, R}} a_1^{p+1} a_2^{r+1} b \leq \frac{b}{(1 - a_1)(1 - a_2)} (a_1^{P+2} a_2 + a_1 a_2^{R+2}).$$

The two monomials on the right have exponents

$$\gamma_b + (P+2)\gamma_1 + \gamma_2 \quad \text{and} \quad \gamma_b + \gamma_1 + (R+2)\gamma_2.$$

If both exponents are strictly larger than $r_\star$, the displayed upper bound is $o(n^{-r_\star})$, proving sufficiency. For necessity, the complement contains $(P+1, 0)$ and $(0, R+1)$. Their monomials have exactly the two displayed exponents. If either strict inequality fails, the corresponding nonnegative monomial is not $o(n^{-r_\star})$, so neither is the entire omitted sum. This proves the claimed equivalence. $\square$

Theorem 2 (thm:transition-mediated-converse-first-order-bracket). *Let $\gamma_j = \alpha_j/(2\alpha_j + d)$ for $j = 1, 2$, $\gamma_b = \beta/(2\beta + d)$, and $\gamma_t = t/(2t + d)$. For every nonempty $\mathcal{M} = \mathcal{M}(\alpha_1, \alpha_2, \beta, t; M, c)$, define the following fixed-proportion*

cross-fitted first-order estimator. Split the sample into three folds F_1, F_2, F_3 whose sizes differ by at most one and, for $v = 1, 2, 3$, cyclically set $\mathcal{T}_v = F_v$, $\mathcal{B}_v = F_{v+1}$, and $\mathcal{E}_v = F_{v+2}$, with indices read modulo three. Let Z_k contain the first k boundary-corrected wavelets from a fixed orthonormal basis with regularity exceeding $\max\{\alpha_1, \alpha_2, \beta, t\}$, and define

$$\text{LS}_{k,[a,b]}(W \mid X, S)(x) = \text{clip}_{[a,b]} \left\{ Z_k(x)^T \left(\sum_i S_i Z_k(X_i) Z_k(X_i)^T \right)^+ \sum_i S_i Z_k(X_i) W_i \right\},$$

where the sums range over the stated training fold and $(\cdot)^+$ denotes the Moore–Penrose inverse. On $\mathcal{T}_v$, with

$$k_{\pi_j} = \max\{1, \lfloor n^{d/(2\alpha_j+d)} \rfloor\}, \quad k_{m_2} = \max\{1, \lfloor n^{d/(2\beta+d)} \rfloor\},$$

fit $\hat{\pi}_{1,v} = \text{LS}_{k_{\pi_1}, [c/2, 1-c/2]}(A_1 \mid H_1, 1)$, $\hat{\pi}_{2,v} = \text{LS}_{k_{\pi_2}, [c/2, 1-c/2]}(A_2 \mid H_2, A_1)$, and $\hat{m}_{2,v} = \text{LS}_{k_{m_2}, [-1, 1]}(Y \mid H_2, A_1 A_2)$. On $\mathcal{B}_v$, form

$$\hat{D}_{2,v} = \hat{m}_{2,v}(H_2) + \frac{A_2}{\hat{\pi}_{2,v}(H_2)} \{Y - \hat{m}_{2,v}(H_2)\}$$

and fit $\hat{m}_{1,v} = \text{LS}_{k_{m_1}, [-L_c, L_c]}(\hat{D}_{2,v} \mid H_1, A_1)$, where $k_{m_1} = \max\{1, \lfloor n^{d/(2t+d)} \rfloor\}$ and $L_c = 1 + 4/c$. Define

$$\hat{\psi}_v^{\text{FO}} = \mathbb{P}_{\mathcal{E}_v} \left[\hat{m}_{1,v}(H_1) + \frac{A_1}{\hat{\pi}_{1,v}(H_1)} \{\hat{D}_{2,v} - \hat{m}_{1,v}(H_1)\} \right], \quad \hat{\psi}^{\text{FO}} = \frac{1}{3} \sum_{v=1}^3 \hat{\psi}_v^{\text{FO}}.$$

Uniformly over $P \in \mathcal{M}$, this estimator satisfies

$$\sup_{P \in \mathcal{M}} E_P[(\hat{\psi}^{\text{FO}} - \psi(P))^2]^{1/2} \lesssim n^{-1/2} + n^{-(\gamma_1 + \gamma_t)} + n^{-(\gamma_2 + \gamma_b)}.$$

If, in addition, $M > 1$ and $0 < \underline{q} < 1 < \bar{q} < \infty$, sufficiently small fixed amplitudes put both fuzzy experiments in $\mathcal{M}$, their product-mixture affinities are bounded below by a constant, and the regular submodel together with the two fuzzy experiments gives

$$\max\{n^{-1/2}, n^{-r_1}, n^{-r_2}\} \lesssim \mathfrak{R}_n(\mathcal{M}) \lesssim n^{-1/2} + n^{-(\gamma_1 + \gamma_t)} + n^{-(\gamma_2 + \gamma_b)}.$$

Thus, under the qualified converse, $\alpha_1 + t \geq d/2$ and $\alpha_2 + \beta \geq d/2$ are necessary for root- n minimax risk, whereas the first-order upper bound makes $\gamma_1 + \gamma_t \geq 1/2$ and $\gamma_2 + \gamma_b \geq 1/2$ sufficient. The n^{-r_1} term is the incoming-transition channel: it is generated by a sign-paired perturbation of π_1 and the incoming-state $C^t(L^1)$ transition law. The n^{-r_2} term is the standard single-time-type terminal channel, and no novelty is asserted for that component. The necessary and sufficient conditions do not coincide in general. Accordingly, this theorem establishes an explicit nonmatching model-specific bracket; it does not determine the minimax rate or a root- n if-and-only-if boundary.

Proof. We prove the first-order upper bound and the qualified converse.

First-order regression rates. By **lem:treated-design-and-backward-regularity**, the regression designs for π_1 , π_2 , and m_2 are uniformly equivalent to Q_1 , Q_2 , and Q_2 , respectively. Conditional on $\mathcal{F}_{T_v}$, the pseudo-outcome is bounded by $L_c = 1 + 4/c$, and the same lemma gives

$$b_{1,v}^\dagger(h_1) = \int \left[\widehat{m}_{2,v}(h_2) + \frac{\pi_2(h_2)}{\widehat{\pi}_{2,v}(h_2)} \{m_2(h_2) - \widehat{m}_{2,v}(h_2)\} \right] g(h_2 | h_1) dh_2$$

with a uniform C^t -norm bound. Apply **lem:bounded-design-wavelet-regression**, conditionally for the random target $b_{1,v}^\dagger$, on folds of size comparable to n . The declared Hölder orders and sieve dimensions yield

$$\begin{aligned} \left\{ E_P \|\widehat{\pi}_{j,v} - \pi_j\|_{L^2(Q_j)}^4 \right\}^{1/4} &\lesssim n^{-\gamma_j}, & \left\{ E_P \|\widehat{m}_{2,v} - m_2\|_{L^2(Q_2)}^4 \right\}^{1/4} &\lesssim n^{-\gamma_b}, \\ \left\{ E_P \|\widehat{m}_{1,v} - b_{1,v}^\dagger\|_{L^2(Q_1)}^4 \right\}^{1/4} &\lesssim n^{-\gamma_t}. \end{aligned}$$

Clipping contracts pointwise distance because each clipping interval contains the corresponding target range.

Apply **lem:direct-bias** fold by fold. Since $1 - w_j = (\widehat{\pi}_j - \pi_j)/\widehat{\pi}_j$ and $\widehat{\pi}_j \geq c/2$, conditional Cauchy–Schwarz gives

$$\begin{aligned} |E_P[\widehat{\psi}_v^{\text{FO}} - \psi | \mathcal{F}_{T_v B_v}]| &\leq \frac{2}{c} \|\widehat{\pi}_{1,v} - \pi_1\|_{L^2(Q_1)} \|\widehat{m}_{1,v} - b_{1,v}^\dagger\|_{L^2(Q_1)} \\ &\quad + \frac{2}{c} \|\widehat{\pi}_{2,v} - \pi_2\|_{L^2(Q_2)} \|\widehat{m}_{2,v} - m_2\|_{L^2(Q_2)}. \end{aligned}$$

Hölder’s inequality over the training folds bounds the $L^2(P)$ -norm of this conditional bias by

$$C\{n^{-(\gamma_1+\gamma_t)} + n^{-(\gamma_2+\gamma_b)}\}.$$

Every evaluation summand is bounded by a constant depending only on c . Conditional independence of the evaluation fold therefore gives

$$E_P[\text{Var}_P(\widehat{\psi}_v^{\text{FO}} | \mathcal{F}_{T_v B_v})] \leq C|\mathcal{E}_v|^{-1} \lesssim n^{-1}.$$

The conditional bias–variance identity and the triangle inequality in $L^2(P)$ prove the asserted risk for each v ; averaging preserves it. This is one sample-measurable estimator, independent of the unknown member P , so it is a valid minimax upper-bound witness.

Feasibility of the fuzzy families. Every transition density satisfies $\sup_{h_1} \|g(\cdot | h_1)\|_{L^1} = 1$; hence $\mathcal{G}_d^t(M)$ is empty when $M < 1$. If $\underline{q} = \bar{q} = 1$, the design condition forces $q_2 = 1$. In a nontrivial first-experiment component with $\lambda = z$, however,

$$\int \frac{z_j u_{j,h}}{1 + 2\eta z_j u_{j,h}} = -2\eta \int \frac{u_{j,h}^2}{1 + 2\eta z_j u_{j,h}} < 0,$$

so the induced Q_2 -density has a nonzero $v(h_2)$ -coefficient. Thus unqualified membership is false.

Now assume $M > 1$ and $0 < q < 1 < \bar{q}$. Disjoint rescaled smooth bumps and denominators uniformly separated from zero satisfy

$$\left\| \sum_j z_j u_{j,h} \right\|_{C^{\alpha_1}} \leq Ch^{-\alpha_1}, \quad \left\| h_1 \mapsto \sum_j \lambda_j \frac{u_{j,h}(h_1)}{1 + 2\eta z_j u_{j,h}(h_1)} v(\cdot) \right\|_{C^t(L^1)} \leq Ch^{-t}.$$

Thus $\eta = c_\eta h^{\alpha_1}$ and $\epsilon = c_\epsilon h^t$, with sufficiently small fixed constants, place $\pi_{1,z}$ and $g_{z,\lambda}$ in their declared balls; $M > 1$ provides slack around the density one. Choose also $\delta > 0$ so that $\delta \|v\|_{C^\beta} \leq M$ and $\delta \|v\|_\infty < 1$. In the second experiment, the same rescaling calculation gives

$$\|\pi_{2,z} - 1/2\|_{C^{\alpha_2}} \leq Cah^{-\alpha_2}, \quad \|m_{2,z,\lambda}\|_{C^\beta} \leq Cb_0 h^{-\beta}.$$

Small amplitude constants give overlap, transition positivity, and valid binary means. The independent-uniform construction gives consistency and both sequential exchangeability conditions, while its specified conditionals give the two state-sufficiency restrictions. Furthermore, mean zero of each bump gives

$$\left| \int \frac{u_{j,h}}{1 + 2\eta z_j u_{j,h}} \right| = 2\eta \left| \int \frac{z_j u_{j,h}^2}{1 + 2\eta z_j u_{j,h}} \right| \leq C\eta h^d.$$

Since $Jh^d \asymp 1$, the first experiment has $\|dQ_2/dh_2 - 1\|_\infty \leq C\eta\epsilon$; both design densities are one in the second experiment. The strict density slack proves membership for all sufficiently large n .

The first fuzzy channel. Relative to the fair-treatment, uniform-transition reference law, the varying part of the one-observation likelihood ratio has disjoint-cell factors

$$f_j = 2\eta(2A_1 - 1)u_{j,h}(H_1), \quad k_j = \epsilon \mathbf{1}\{A_1 = 1\}u_{j,h}(H_1)v(H_2).$$

Mean zero of u and v gives $E_0 f_j = E_0 k_j = E_0(f_j k_j) = 0$. The conditional squared norms within a cell are at most $C\eta^2$ and $C\epsilon^2$, and each cell has probability comparable to J^{-1} . Hence `lem:symmetric-cell-mixture-affinity` gives

$$H^2(\bar{P}_+, \bar{P}_-) \leq Cn^2 \eta^2 \epsilon^2 / J.$$

Writing $I_v = \int v^2 > 0$, direct evaluation of the g-formula target gives

$$\psi_{z,\lambda} = \delta\epsilon I_v \sum_j \lambda_j \int \frac{u_{j,h}}{1 + 2\eta z_j u_{j,h}}.$$

For $\lambda = z$, the displayed mean-zero identity makes every summand negative; for $\lambda = -z$, every summand is positive. Uniform denominator bounds and $\sum_j \int u_{j,h}^2 \asymp 1$ separate the two prior supports from zero by at least $c_0 \eta \epsilon$. With

$$h \asymp n^{-2/\{2(\alpha_1+t)+d\}},$$

we have $n^2\eta^2\epsilon^2/J \asymp c_\eta^2 c_\epsilon^2$ and $\eta\epsilon \asymp n^{-r_1}$. Reducing the fixed amplitude constants makes $1 - H^2(\bar{P}_+, \bar{P}_-) \geq \rho$ for some $\rho > 0$.

The second fuzzy channel. The varying likelihood factors are

$$f_j = 2a\mathbf{1}\{A_1 = 1\}(2A_2 - 1)v_{j,h}(H_2), \quad k_j = b_0\mathbf{1}\{A_1 = A_2 = 1\}Yv_{j,h}(H_2).$$

The same orthogonality calculation and symmetric-cell lemma give

$$H^2(\bar{P}_+, \bar{P}_-) \leq Cn^2a^2b_0^2/J.$$

Here

$$\psi_{z,\lambda} = b_0 \sum_j \lambda_j \int \frac{v_{j,h}}{1 + 2az_j v_{j,h}}.$$

Mean zero gives opposite signs under $\lambda = z$ and $\lambda = -z$, with separation from zero at least c_1ab_0 . Taking $h \asymp n^{-2/\{2(\alpha_2+\beta)+d\}}$ and sufficiently small amplitudes gives positive affinity and $ab_0 \asymp n^{-r_2}$.

For either experiment, let the two target supports lie in $(-\infty, -\Delta]$ and $[\Delta, \infty)$. Threshold any estimator at zero. On a testing error its absolute estimation error is at least Δ , whence

$$\sup_{P \in \mathcal{M}} E_P[(\hat{\psi} - \psi(P))^2] \geq \frac{\Delta^2}{2} \{1 - \text{TV}(\bar{P}_+, \bar{P}_-)\}.$$

If $A = \int \sqrt{d\bar{P}_+ d\bar{P}_-} = 1 - H^2(\bar{P}_+, \bar{P}_-)$, Cauchy–Schwarz gives

$$\text{TV}(\bar{P}_+, \bar{P}_-) \leq \frac{1}{2} \left\{ \int (\sqrt{p} - \sqrt{q})^2 \int (\sqrt{p} + \sqrt{q})^2 \right\}^{1/2} = \sqrt{1 - A^2}.$$

Thus $A \geq \rho$ makes the testing factor uniformly positive and yields the n^{-r_1} and n^{-r_2} RMSE floors.

In the regular submodel take $\vartheta_\pm = \pm\kappa n^{-1/2}$. The target equals ϑ . Only the stratum $A_1 = A_2 = 1$, of probability $1/4$, varies, and the conditional Bernoulli affinity in that stratum is $\sqrt{1 - \kappa^2/n}$. Hence the one-observation affinity is

$$A(P_{\vartheta_+}, P_{\vartheta_-}) = \frac{3}{4} + \frac{1}{4} \sqrt{1 - \kappa^2/n} \geq 1 - \frac{\kappa^2}{4n}.$$

Affinity multiplies under independent products, so for all sufficiently large n , $A(P_{\vartheta_+}^{\otimes n}, P_{\vartheta_-}^{\otimes n}) \geq e^{-\kappa^2/2} > 0$. The same testing calculation and target separation $2\kappa n^{-1/2}$ give the parametric floor. Combining the three experiments proves the lower bound. Finally, $2s/(2s + d) \geq 1/2$ exactly when $s \geq d/2$, which gives the stated necessary elbow conditions; the first-order upper bound gives the stated sufficient conditions. $\square$

Justification. This is the main statistical result: one explicit cross-fitted estimator, two fuzzy nonparametric floors, and a parametric floor. *Gap.* The lower and upper exponents differ in general, so the minimax rate and sharp root- n boundary remain open. *Consumer.* It supplies an actionable model-specific bracket for the non-root- n time-varying-treatment problem.

7 Interpretation

Transition smoothness can enter the first decision’s difficulty at first order. The lower experiment changes π_1 and the incoming-state behavior of g with paired signs, producing target separation of order n^{-r_1} while keeping mixture affinity positive. If $g(h_2 \mid h_1) = g_0(h_2)$, then m_1 is constant and the target has no nonconstant incoming-state component. A finite-smoothness analytic-outgoing subclass shows that the structural channel can be inactive at the necessary elbow for every $\alpha_1 > 0$ when $t_0 > d/2$, but the note does not turn that necessary condition into a root- n upper theorem. The auxiliary monotone-MAR law is only a comparator representation of observed treated-regime coordinates, not a causal interpretation of L_1 .

Technical internal limitation (diagnostic only)

The proved estimator is first order. Its upper exponents are products of ordinary nuisance-regression rates, whereas the fuzzy converses depend on sums of structural smoothness orders. The projected higher-order construction is diagnostic rather than attaining: empirical Gram rank, stochastic higher-order chains, and the transported mixed remainder are not resolved into a matching upper bound. The auxiliary comparator now has a complete outcome kernel and explicit MAR identities, but it remains a separate statistical construction from the causal full-data extension and must not be read as identifying L_1 with $H_2(1)$.

8 Honest scope

The formal results are conditional on the displayed causal, state-sufficiency, overlap, design-density, and Hölder conditions. The upper bound holds for every nonempty declared model. The fuzzy converse additionally uses $M > 1$, strict design-density slack around one, and sufficiently small fixed amplitudes. The rough-subclass proposition proves an n^{-r_1} floor on one fixed n -independent subclass and a precise noncoverage result for the fully specified auxiliary monotone-MAR laws induced from it. Equality of observed coarsenings is asserted only on the stated observed events, and no broad-model state-consistency condition is imposed. The note does not prove the conjectured matched rate, a sharp elbow, root- n estimability on the analytic-outgoing subclass, reverse comparator inclusion, broad literature separation, or priority.

References

- [1] Carlos Cinelli, Avi Feller, Guido Imbens, Edward H. Kennedy, Sara Magliacane, and Jose Zubizarreta (2025). Challenges in Statistics: A Dozen Challenges in Causality and Causal Inference. arXiv:2508.17099.

- [2] Alexander R. Luedtke, Oleg Sofrygin, Mark J. van der Laan, and Marco Carone (2017). Sequential Double Robustness in Right-Censored Longitudinal Models. arXiv:1705.02459.
- [3] James M. Robins, Lingling Li, Eric Tchetgen Tchetgen, and Aad van der Vaart (2016). Technical Report: Higher Order Influence Functions and Minimax Estimation of Nonlinear Functionals. arXiv:1601.05820, Section 7.2.2, Theorems 48–49.
- [4] James M. Robins, Lingling Li, Eric Tchetgen Tchetgen, and Aad van der Vaart (2016). Higher order influence functions and minimax estimation of nonlinear functionals. *Stochastic Processes and their Applications* 126, 2547–2571.
- [5] James M. Robins, Eric Tchetgen Tchetgen, Lingling Li, and Aad van der Vaart (2009). Semiparametric minimax rates. *Electronic Journal of Statistics* 3, 1305–1321.
- [6] Matteo Bonvini, Edward H. Kennedy, and Luke Keele (2025). Higher-order estimators of time-varying effects in anisotropic smoothness models. Unpublished work in progress disclosed in the EcoSta program as Abstract A0576; no public manuscript is used in this note.
- [7] Miguel A. Hernán and James M. Robins (2020). *Causal Inference: What If*. Chapman and Hall/CRC.
- [8] Alexandre B. Tsybakov (2009). *Introduction to Nonparametric Estimation*. Springer.